

Final Term

Unequal Probability Sampling

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- The expected value of HH estimator is

$$E(y'_{HH}) = \sum y'_{HH} p_i p_j = 7 = Y$$

$$Var(y'_{PPS}) = E(y'^2_{PPS}) - Y^2$$

$$Var(y'_{PPS}) = 49.5 - 49 = 0.5$$

➤ Using the Formula

most imp formula

$$Var(y'_{PPS}) = \frac{1}{n} \left[\sum_{i=1}^N \frac{Y_i^2}{P_i} - Y^2 \right] = \frac{1}{2} [50 - 49] = 0.50$$

Y_i

Z_i

$$P_i = Z_i / \sum Z_i$$

0.5	1	0.1
1.2	2	0.2
2.1	3	0.3
3.2	4	0.4

Q. Describe the Lahiri's Method of selection?

Answer:

➤ Lahiri's Method of Selection

- A pair of random numbers is chosen such that one from 1 to N and other 1 to Z_{max} (say R)

imp formulas

$$P_i(ith) = \frac{1}{N} [Z_i / Z_{\max}]$$

- if R exceeds the size of the ith unit; then that unit is rejected otherwise it is accepted.
- probability of selecting the ith unit at the first draw.

Sr.No	Y _i	Z _i
1	0.5	15
2	1.2	20
3	2.1	7

4 3.2 13

Q. Example for Lahiri's Method

S.No. of Village	No. of Trees Y_i	Area in Acres Z_i	P_i	S.No. of Village	No. of Trees Y_i	Area in Acres Z_i	P_i
1	2328	328	0.700	11	2483	537	0.120
2	754	80	0.020	12	128	7	0.002
3	105	6	0.001	13	102	8	0.002
4	949	156	0.030	14	60	22	0.005
5	3091	428	0.100	15	0	4	0.001
6	1736	401	0.090	16	11799	1234	0.280
7	840	94	0.020	17	26	3	0.001
8	311	63	0.010	18	317	30	0.010
9	0	51	0.010	19	190	40	0.010
10	3044	946	0.210	20	180	20	0.004

➤ **Selection with Lahiri's Method**

Pair of Random Numbers	No. of Village	Y_i	P_i	$\frac{y_i}{p_i}$
(16,37)	16	11799	0.28	42139.286
(5,38)	5	3091	0.10	30910.00
(16,23)	16	11799	0.28	42139.286
(10,54)	10	3044	0.21	14495.238
(10,54)	10	3044	0.21	14495.238
				144179.05

➤ **The Estimated Total**

- The estimated value for population total using Lahiri's method of selection

$$y'_{PPS} = \frac{1}{n} \sum_{i=1}^5 \frac{y_i}{p_i} = \frac{144179.05}{5} = 28836$$

Q. Explain the concept of Unequal Probability Sampling Without Replacement?

Answer:

- let a sample of two units is selected from a population of N units.

- Let the probability of the selection of the i th unit is $P_i = Z_i/Z$. imp mcq's
- Suppose the i th unit is not selected at the first draw but the j th unit is selected ($j \neq i$) then the probability of selecting the j th unit at the first draw is $P_j = Z_j/Z$;
- The conditional probability of selecting the i th unit at the second draw is $Z_i/(Z - Z_j)$ imp mcq
- The probability of inclusion of i th unit at the second draw to be included in the sample is the sum of the product that the j th unit is selected at the first draw and the i th unit is selected at the second draw given the j th unit is selected at the first draw i.e.

$$\sum_{j \neq i}^N P_j \frac{P_i}{1 - P_j}$$

- The total probability π_i , the probability of inclusion of the i th population unit to be in the sample is

$$\begin{aligned} \pi_i &= P_i + \sum_{j \neq i}^N P_j \frac{P_i}{1 - P_j} \\ &= P_i \left[1 + \sum_{j \neq i}^N \frac{P_j}{1 - P_j} - \frac{P_i}{1 - P_i} \right] \end{aligned}$$

- The probability that both i th and j th units are in the sample is denoted by π_{ij} and is defined as

$$\begin{aligned} \pi_{ij} &= P_i P_{j|i} + P_j P_{i|j} = P_i \frac{P_j}{1 - P_i} + P_j \frac{P_i}{1 - P_j} \\ &= P_i P_j \left[\frac{1}{1 - P_i} + \frac{1}{1 - P_j} \right] \end{aligned}$$

Q. The Horvitz Thompson Estimator?

most imp short question

Answer:

➤ HT Estimator

- The general theory of unequal probabilities sampling without replacement was presented firstly by Horvitz and Thompson (1952).
- An unbiased estimator suggested by them for population total Y is

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i} = \sum_{i=1}^N a_i \frac{Y_i}{\pi_i},$$

most imp short question

Q. Prove that HT Estimator is unbiased estimator.

➤ The Horvitz and Thompson in 1952 suggested the estimator total is

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i} = \sum_{i=1}^N a_i \frac{Y_i}{\pi_i},$$

$$E(y'_{HT}) = \sum_{i=1}^N \pi_i \frac{Y_i}{\pi_i} = \sum_{i=1}^N Y_i = Y$$

Q.81 what are the Some Relations of π_i ?**Answer:**➤ **Relations of π_i**

most imp short question

• The following are some relations of

3 marks

i. $\sum_{i=1}^N \pi_i = n$

ii. $\sum_{j \neq i}^N \pi_{ij} = (n-1) \pi_i$

iii. $\sum_{j \neq i}^N (\pi_{ij} - \pi_i \pi_j) = -\pi_i (1 - \pi_i)$

iv. $\sum_{i=1}^N \sum_{j \neq i}^N \pi_i \pi_j = n \sum_{i=1}^N \pi_i - \sum_{i=1}^N \pi_i^2$

imp mcq's

➤ **Relation (i)**• We know $\sum_{i=1}^N a_i = n$ • Taking expectation $\sum_{i=1}^N \pi_i = n$ ➤ **Relation (ii)**

$$\sum_{j \neq i}^N \pi_{ij} = (n-1) \pi_i$$

• is sum of all the probabilities of the samples containing ith and jth units

- is the sum of the probabilities containing first and second units; first and third units; and so on.
- Thus every $P(s)$ containing the first unit occurs $(n-1)$ times in this sum as the sample has $(n-1)$ other members in it and it occurs once for each of these members.

$$\sum_{j \neq i}^N \pi_{ij}, \sum_{j \neq i}^N \pi_{ij} = (n-1)\pi_i$$

➤ **Relation (iii)**

$$\sum_{j \neq i}^N (\pi_{ij} - \pi_i \pi_j) = -\pi_i(1 - \pi_i)$$

Taking L.H.S

$$\sum_{j \neq i}^N (\pi_{ij} - \pi_i \pi_j) = \sum_{j \neq i}^N \pi_{ij} - \pi_i \sum_{j \neq i}^N \pi_j$$

$$\sum_{j \neq i}^N \pi_{ij} - \pi_i (\sum_{j=1}^N \pi_j - \pi_i)$$

$$\sum_{j \neq i}^N \pi_{ij} - \pi_i (\sum_{j=1}^N \pi_j - \pi_i)$$

Using (i) and (ii) relation

$$= (n-1)\pi_i - \pi_i(n - \pi_i) \\ = -\pi_i(1 - \pi_i)$$

➤ **Relation (iv)**

$$\sum_{i=1}^N \sum_{j \neq i}^N \pi_i \pi_j = n \sum_{i=1}^N \pi_i - \sum_{i=1}^N \pi_i^2$$

$$\sum_{i=1}^N \sum_{j \neq i}^N \pi_i \pi_j = \sum_{i=1}^N \pi_i \sum_{j \neq i}^N \pi_j$$

Q.82 Find the variance expression of Horvitz Thompson estimator?

Answer:

➤ **HT Estimator**

only formula mcq's

- The Horvitz and Thompson in 1952 suggested the estimator total is

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i} = \sum_{i=1}^N a_i \frac{Y_i}{\pi_i},$$

$$E(y'_{HT}) = \sum_{i=1}^N \pi_i \frac{Y_i}{\pi_i} = \sum_{i=1}^N Y_i = Y$$

➤ **Variance of HT Estimator**

$$Var(y'_{HT}) = E(y'_{HT})^2 - [E(y'_{HT})]^2$$

$$\begin{aligned}
 y'_{HT} &= \sum_{i=1}^n \frac{y_i}{\pi_i} = \sum_{i=1}^N a_i \frac{Y_i}{\pi_i}, \\
 \text{Var}(y'_{HT}) &= E \left(\sum_{i=1}^N a_i \frac{Y_i}{\pi_i} \right)^2 \\
 &\quad - \left[E \left(\sum_{i=1}^N a_i \frac{Y_i}{\pi_i} \right) \right]^2 \\
 \text{Var}(y'_{HT}) &= E \left(\sum_{i=1}^N a_i \frac{Y_i}{\pi_i} \right)^2 \\
 &\quad - \left[\left(\sum_{i=1}^N E(a_i) \frac{Y_i}{\pi_i} \right) \right]^2 \\
 E \left(\sum_{i=1}^N a_i \frac{Y_i}{\pi_i} \right)^2 &= \left[\sum_{i=1}^N E(a_i^2) \frac{Y_i^2}{\pi_i^2} \right] \\
 &\quad + \left[\sum_{\substack{i,j \\ j \neq i}}^N \sum E(a_i a_j) \frac{Y_i}{\pi_i} \frac{Y_j}{\pi_j} \right] \\
 \text{Var}(y'_{HT}) &= E \left(\sum_{i=1}^N a_i \frac{Y_i}{\pi_i} \right)^2 \\
 &\quad - \left[\left(\sum_{i=1}^N E(a_i) \frac{Y_i}{\pi_i} \right) \right]^2 \\
 &\quad \left[\sum_{i=1}^N E(a_i^2) \frac{Y_i^2}{\pi_i^2} \right] + \left[\sum_{\substack{i,j \\ j \neq i}}^N \sum E(a_i a_j) \frac{Y_i}{\pi_i} \frac{Y_j}{\pi_j} \right] \\
 &\quad - \left[\sum_{i=1}^N [E(a_i)]^2 \frac{Y_i^2}{\pi_i^2} \right] - \sum_{\substack{i,j \\ j \neq i}}^N \sum E(a_i) E(a_j) \frac{Y_i}{\pi_i} \frac{Y_j}{\pi_j} \\
 &= \sum_{i=1}^N \frac{Y_i^2}{\pi_i^2} \left[E(a_i^2) - (E(a_i))^2 \right] \\
 &\quad - \sum_{\substack{i,j=1 \\ j \neq i}}^N \frac{Y_i Y_j}{\pi_i \pi_j} \left[E(a_i a_j) - E(a_i) E(a_j) \right] \\
 &= \sum_{i=1}^N \frac{Y_i^2}{\pi_i^2} \left[E(a_i^2) - (E(a_i))^2 \right] \\
 &\quad - \sum_{\substack{i,j=1 \\ j \neq i}}^N \frac{Y_i Y_j}{\pi_i \pi_j} \left[E(a_i a_j) - E(a_i) E(a_j) \right]
 \end{aligned}$$

$$= \sum_{i=1}^N \frac{Y_i^2}{\pi_i^2} \left[E(a_i^2) - (E(a_i))^2 \right] \\ - \sum_{\substack{i,j=1 \\ j \neq i}}^N \frac{Y_i Y_j}{\pi_i \pi_j} \left[E(a_i a_j) - E(a_i) E(a_j) \right]$$

$$\text{Var}(a_i) = \pi_i (1 - \pi_i)$$

$$\text{Cov}(a_i, a_j) = \pi_{ij} - \pi_i \pi_j$$

$$\text{Var}(y'_{HT}) = \sum_{i=1}^N \frac{Y_i^2}{\pi_i^2} \text{Var}(a_i) + \\ \sum_{\substack{i,j=1 \\ j \neq i}}^N \frac{Y_i Y_j}{\pi_i \pi_j} \text{Cov}(a_i, a_j)$$

$$\text{Var}(y'_{HT}) = \sum_{i=1}^N (1 - \pi_i) \pi_i \frac{Y_i^2}{\pi_i^2} \\ + \sum_{\substack{i,j=1 \\ j \neq i}}^N (\pi_{ij} - \pi_i \pi_j) \frac{Y_i Y_j}{\pi_i \pi_j}$$

$$\text{Var}(y'_{HT}) = \sum_{i=1}^N (1 - \pi_i) \pi_i \frac{Y_i^2}{\pi_i^2} \\ + \sum_{\substack{i,j=1 \\ j \neq i}}^N (\pi_{ij} - \pi_i \pi_j) \frac{Y_i Y_j}{\pi_i \pi_j}$$

$$\sum_{j \neq i} (\pi_{ij} - \pi_i \pi_j) = -\pi_i (1 - \pi_i)$$

$$\text{Var}(y'_{HT}) = \sum_{i=1}^N (1 - \pi_i) \pi_i \frac{Y_i^2}{\pi_i^2} \\ + \sum_{\substack{i,j=1 \\ j \neq i}}^N (\pi_{ij} - \pi_i \pi_j) \frac{Y_i Y_j}{\pi_i \pi_j}$$

$$\text{Var}(y'_{HT}) = \sum_{\substack{i=1 \\ j \neq i}}^N \sum (\pi_i \pi_j - \pi_{ij}) \frac{Y_i^2}{\pi_i^2} \\ + \sum_{\substack{i,j=1 \\ j \neq i}}^N \frac{\pi_{ij} - \pi_i \pi_j}{\pi_i \pi_j} Y_i Y_j$$

$$\begin{aligned}
\text{Var}(y'_{HT}) &= \sum_{i=1}^N \sum_{j \neq i}^N (\pi_i \pi_j - \pi_{ij}) \frac{Y_i^2}{\pi_i^2} \\
&+ \sum_{i,j=1}^N \sum_{j \neq i}^N \frac{\pi_{ij} - \pi_i \pi_j}{\pi_i \pi_j} Y_i Y_j \\
&\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i^2}{\pi_i^2} + \frac{Y_j^2}{\pi_j^2} \right) \\
&- \sum_{i,j=1}^N \sum_{j \neq i}^N \frac{\pi_i \pi_j - \pi_{ij}}{\pi_i \pi_j} Y_i Y_j \\
&\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i^2}{\pi_i^2} + \frac{Y_j^2}{\pi_j^2} \right) \\
&- \sum_{i,j=1}^N \sum_{j \neq i}^N \frac{\pi_i \pi_j - \pi_{ij}}{\pi_i \pi_j} Y_i Y_j \\
&\frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N (\pi_i \pi_j - \pi_{ij}) \\
&\left(\frac{Y_i^2}{\pi_i^2} + \frac{Y_j^2}{\pi_j^2} - 2 \frac{Y_i Y_j}{\pi_i \pi_j} \right) \\
\text{Var}_{\text{SYG}}(y'_{HT}) &= \frac{1}{2} \sum_{i,j}^N \sum_{j \neq i}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i}{\pi_i} - \frac{Y_j}{\pi_j} \right)^2
\end{aligned}$$

formula short Q

Q. Calculate HT estimator by taking all possible samples of size 2 without replacement. Further find mean and variance.

Ans:

Small Population Example

- The following is a population with four values with respective size.

Unit No.	1	2	3	4
Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4
P_i	0.1	0.2	0.3	0.4

- We are interested to take all possible samples of size 2.

- HT estimator will be calculated for all the samples. Further mean and variance will be obtained.

Unit No.	1	2	3	4
Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4
P_i	0.1	0.2	0.3	0.4

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i}$$

$$P_i = Z_i / \sum Z_i$$

$$y'_{HT} = \frac{y_1}{\pi_1} + \frac{y_2}{\pi_2}$$

$$\pi_i = P_i \left[1 + \sum_{j=1}^N \frac{P_j}{1-P_j} - \frac{P_i}{1-P_i} \right]$$

$$\pi_1 = P_1 \left[1 + \sum_{j=1}^N \frac{P_j}{1-P_j} - \frac{P_1}{1-P_1} \right]$$

$$\sum \frac{P_j}{1-P_j} = 1.456$$

$$\pi_1 = 0.1(1 + 1.456 - 0.111)$$

$$= 0.2345$$

Unit No.	1	2	3	4
Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4
P_i	0.1	0.2	0.3	0.4
π_i	0.2345	0.4413	0.6083	0.7159

$$\sum \frac{P_j}{1-P_j} = 1.456$$

$$\pi_1 = 0.1(1 + 1.456 - 0.111)$$

$$= 0.2345$$

$$\pi_{ij} = P_i P_j \left[\frac{1}{1-P_i} + \frac{1}{1-P_j} \right]$$

$$\pi_{ij} = 0.1 * 0.2 \left[\frac{1}{0.9} + \frac{1}{0.8} \right]$$

$$\pi_{ij} = 0.1 * 0.2 [1.111 + 1.25]$$

$$= 0.0472$$

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i}$$

$$E(y'_{HT}) = 7.0000$$

Table given hota ha mean or variance find krny hoty
feb 2024

Unit	y_i	y_j	π_i	π_j	π_{ij}	y'_{HT}	$\pi_{ij} y'_{HT}$	$\pi_{ij} (y'_{HT})^2$
1, 2	0.5	1.2	0.2345	0.4413	0.0472	4.8514	0.2291	1.1114
1, 3	0.5	2.1	0.2345	0.6083	0.0762	5.5840	0.4255	2.3757
1, 4	0.5	3.2	0.2345	0.7159	0.1111	6.6020	0.7336	4.8430
2, 3	1.2	2.1	0.4413	0.6083	0.1607	6.1715	0.9918	6.1212
2, 4	1.2	3.2	0.4413	0.7159	0.2333	7.1895	1.6775	12.0607
3, 4	2.1	3.2	0.6083	0.7159	0.3714	7.9221	2.9425	23.3109
					1.00		7.00	49.823

long question

$$Var(y'_{HT}) = 49.8229 - (7.0000)^2$$

$$= 0.8229$$

Q.85 what are the HT Estimator for with Replacement Sampling?

Answer:

➤ **Sample of Four Units**

Unit No.	1	2	3	4
Y_i	60	60	14	1
P_i	0.05	0.05	0.02	0.01

- Y_i is the count of animals from the sample of four strips.

➤ **HH Estimator**

Unit No.	1	2	3	4
----------	---	---	---	---

Y_i	60	60	14	1
P_i	0.05	0.05	0.02	0.01

$$= \frac{1}{4} \left(\frac{60}{0.05} + \frac{60}{0.05} + \frac{14}{0.02} + \frac{1}{0.01} \right)$$

$$= 800$$

➤ HT Estimator

Unit No.	1	2	3	4
Y_i	60	60	14	1
P_i	0.05	0.05	0.02	0.01

$$\pi_i = 1 - (1 - p_i)^n$$

$$y'_{HT} = \sum_{i=1}^n \frac{y_i}{\pi_i}$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= P(A) + P(B) - [1 - P(A^c \cap B^c)]$$

$$\pi_{ij} = \pi_i + \pi_j - [1 - (1 - p_i - p_j)^n]$$

Q. Perform PPS Sampling With Replacement using the following data in R?

Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4

Answer:

```
y<-c(0.5,1.2,2.1,3.2)
zi<-c(1,2,3,4)
z<-sum(zi)
pi<-zi/z
N<-4;n=2;means=c();
#----With PPS-----
for(i in 1:10)
{s <- sample(1:N,n,replace=TRUE,prob=pi)
means[i]<-mean(y[s]) }
```

imp s/Q

Repeated Question in past paper

mean(means)

Q. Select 10000 samples with PPS Sampling With Replacement using the following data in R. Find mean of each sample. Further find mean of means

Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4

```

y<-c(0.5,1.2,2.1,3.2)
zi<-c(1,2,3,4)
z<-sum(zi)
pi<-zi/z
N<-4;n=2
#----With PPS-----
for(i in 1:10000)
{s <- sample(1:N,n,replace=TRUE,prob=pi)
means[i]<-mean(y[s]) }
mean(means)
    
```

imp s/q

Q. Select 10000 samples with PPS Sampling With Replacement using the following data in R. Find HH estimator from each sample. Further find mean of HH estimators.

Y_i	0.5	1.2	2.1	3.2
Z_i	1	2	3	4

Answer:

```

y<-c(0.5,1.2,2.1,3.2)
zi<-c(1,2,3,4);z<-sum(zi);pi<-zi/z
N<-4;n=2
for(i in 1:10000)
{s <- sample(1:N,n,replace=TRUE,prob=pi)
hh[i] <- mean(y[s]/pi[s])/N }
mean(hh)
    
```

s/q

Example:

Answer:

➤ **Population from R**

- Read in the trees data set from R
- let the variable of interest y be tree volume and draw by draw selection probability be proportional to girth.
- The variable of interest is tree volume.

```
# -----Reading the data-----
y <- trees$Volume
zi <- trees$Girth
y
10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 24.2 21.0 21.4 21.3 19.1 22.2 33.8 27.4
25.7 24.9 34.5 31.7 36.3 38.3 42.6 55.4 55.7 58.3 51.5 51.0 77.0
zi
8.3 8.6 8.8 10.5 10.7 10.8 11.0 11.0 11.1 11.2 11.3 11.4 11.4 11.7 12.0 12.9 12.9 13.3
13.7 13.8 14.0 14.2 14.5 16.0 16.3 17.3 17.5 17.9 18.0 18.0 20.6
```

➤ Defining Terms

```
# -----Defining Population----
y <- trees$Volume
zi <- trees$Girth
pi <- zi/sum(zi)
pi
0.02020940 0.02093986 0.02142683 0.02556611 0.02605308 0.02629657 0.02678354
0.02678354 0.02702703 0.02727051 0.02751400 0.02775749 0.02775749 0.02848795
0.02921841 0.03140979 0.03140979 0.03238374 0.03335768 0.03360117 0.03408814
0.03457512 0.03530558 0.0389578 0.03968834 0.04212320 0.04261018 0.04358412
0.04382761 0.04382761 0.05015827
sum(zi)
410.7
```

➤ Sampling With PPS

```
y <- trees$Volume
zi <- trees$Girth
pi <- zi/sum(zi)
N <- 31; n = 10;
#----Sampling with PPS----
s <- sample(1:N, n, replace=TRUE, prob=pi)
y[s]
mean(y[s])
mu <- mean(y)
```

➤ With SRS

```
y <- trees$Volume
zi <- trees$Girth
pi <- zi/sum(zi)
```



```

N<-31;n=10;
#----Sampling with PPS----
s <- sample(1:N,n,replace=TRUE,prob=NULL)
y[s]
mean(y[s])
mu <- mean(y)
> mean(y[s])
[1] 37.96
mu
[1] 30.17

```

➤ Simulation with PPS

```

y <- trees$Volume
zi <- trees$Girth
pi<-zi/sum(zi)
N<-31;n=10;
#----With PPS-----
for(i in 1:10000)
{ s <- sample(1:N,n,replace=TRUE,prob=pi)
means[i]<-mean(y[s]) }
mean(means)
mean
33.37

```

➤ Simulation with SRSWR

```

y <- trees$Volume
zi <- trees$Girth
pi<-zi/sum(zi)
N<-31;n=10;
#----SRSWR----
for(i in 1:10000)
{ s <- sample(1:N,n,replace=TRUE,prob=NULL)
means[i]<-mean(y[s]) }
mean(means)
mean
30.25

```

➤ Observations

- Mean of Population is 30.17
- Mean of means with SRS is 30.25-
- Mean of means with PPS is 33.37

Q. Read the data from the following command and select 10000 samples with PPS Sampling with replacement using the following data in R. Find HH estimator from each sample. Further find mean of HH estimators.

```

y <- trees$Volume
zi <- trees$Girth

```


Answer:

➤ **Population from R**

-----Reading the data-----

y <- trees\$Volume

zi <- trees\$Girth

y

10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 24.2 21.0 21.4 21.3 19.1 22.2 33.8 27.4 25.7
24.9 34.5 31.7 36.3 38.3 42.6 55.4 55.7 58.3 51.5 51.0 77.0

➤ **Population from R**

y

10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 24.2 21.0 21.4 21.3 19.1 22.2 33.8 27.4 25.7
24.9 34.5 31.7 36.3 38.3 42.6 55.4 55.7 58.3 51.5 51.0 77.0

zi

8.3 8.6 8.8 10.5 10.7 10.8 11.0 11.0 11.1 11.2 11.3 11.4 11.4 11.7 12.0 12.9 12.9 13.3 13.7
13.8 14.0 14.2 14.5 16.0 16.3 17.3 17.5 17.9 18.0 18.0 20.6

y <- trees\$Volume

zi <- trees\$Girth

pi <- zi/sum(zi); N <- 31; n = 10;

for(i in 1:10000)

{ s <- sample(1:N, n, replace=TRUE, prob=pi)

hh[i] <- mean(y[s]/pi[s])/N }

mean(hh)

mean(hh)

[1] 30.18311

program only

Q. Select 10000 samples with PPS Sampling with replacement using the following data in R. Find HT estimator from each sample. Further find mean of HT estimators.

Y _i	0.5	1.2	2.1	3.2
Z _i	1	2	3	4

Answer:

-----Defining Population-----

y <- c(0.5, 1.2, 2.1, 3.2)

zi <- c(1, 2, 3, 4)

z <- sum(zi)

pi <- zi/z; N = 4; n = 2

---Calculation of Pi---

s/q


```
pii <- 1 - (1-pi)^n
```

```
#---Looping-----
```

```
for(i in 1:10000)
```

```
{s <- sample(1:N,n,replace=TRUE,prob=pi)
```

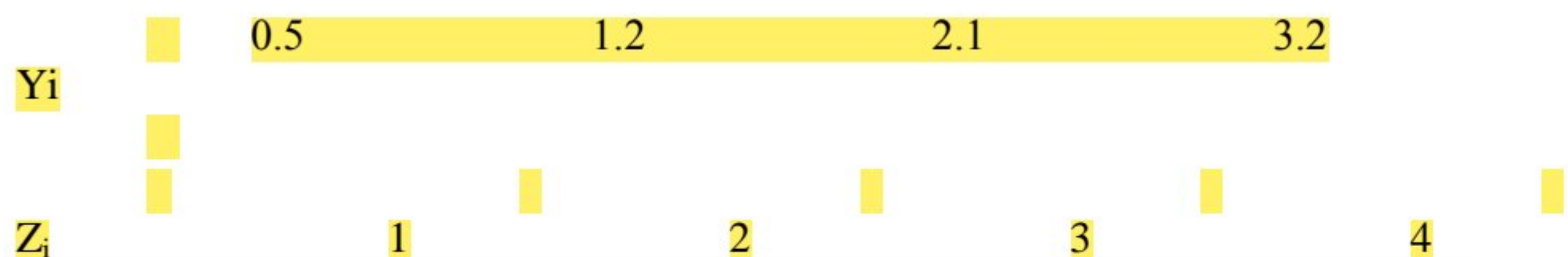
```
yu=unique(s)
```

```
ht[i] <- sum(y[yu]/pii[yu])/N}
```

```
mean(ht)
```

All imp

Q. Select 10000 samples with PPS Sampling without replacement using the following data in R. Find HT estimator from each sample. Further find mean of HT estimators.



Answer:

```
# -----Defining Population----
```

```
y<-c(0.5,1.2,2.1,3.2)
```

```
zi<-c(1,2,3,4)
```

```
z<-sum(zi)
```

```
pi<-zi/z;N=4;n=2
```

```
#---Calculation of Pi----
```

```
piinv<-1-pi
```

```
sm<-sum(pi/piinv)
```

```
pii <- pi*(1+sm-pi/(1-pi))
```

s/q

➤ **ith HT Estimator**

```
#---Looping-----
```

```
hht=c();
```

```
for(i in 1:10000)
```

```
{s <- sample(1:N,n,replace=FALSE,prob=pi)
```

```
hht[i] <- sum(y[s]/pii[s])/N}
```

```
mean(hht)
```

Q. Read the data from the following command and select 10000 samples with PPS Sampling with replacement using the following data in R. Find HT estimator from each sample. Further find mean of HT estimators.

```
y <- trees$Volume
```

```
zi <- trees$Girth
```

Answer:

```
# -----Reading the data-----
```



```
y <- trees$Volume
zi <- trees$Girth
y
10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 24.2 21.0 21.4 21.3 19.1 22.2 33.8 27.4 25.7
24.9 34.5 31.7 36.3 38.3 42.6 55.4 55.7 58.3 51.5 51.0 77.0
y
10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 24.2 21.0 21.4 21.3 19.1 22.2 33.8 27.4 25.7
24.9 34.5 31.7 36.3 38.3 42.6 55.4 55.7 58.3 51.5 51.0 77.0
zi
8.3 8.6 8.8 10.5 10.7 10.8 11.0 11.0 11.1 11.2 11.3 11.4 11.4 11.7 12.0 12.9 12.9 13.3 13.7
13.8 14.0 14.2 14.5 16.0 16.3 17.3 17.5 17.9 18.0 18.0 20.6
sum(zi)
410.7
```

-----Defining Population-----

```
y <- trees$Volume
zi <- trees$Girth
pi <- zi/sum(zi)
pi
0.02020940 0.02093986 0.02142683 0.02556611 0.02605308 0.02629657 0.02678354
0.02678354 0.02702703 0.02727051 0.02751400 0.02775749 0.02775749 0.02848795
0.02921841 0.03140979 0.03140979 0.03238374 0.03335768 0.03360117 0.03408814
0.03457512 0.03530558 0.0389578 0.03968834 0.04212320 0.04261018 0.04358412
0.04382761 0.04382761 0.05015827
```

```
y <- trees$Volume
zi <- trees$Girth
pi <- zi/sum(zi)
N<-31;n=10;
#---Calculation of Pi---
pii <- 1 - (1-pi)^n
pii
```

➤ **Values of pii**

```
> pii
0.1846714 0.1907295 0.1947457 0.2281662 0.2320147 0.2339325 0.2377552
0.2377552 0.2396601 0.2415607 0.2434571 0.2453491 0.2453491 0.2509998
0.2566124 0.2732237 0.2732237 0.2804987 0.2877080 0.2895002 0.2930723
0.2966283 0.3019321 0.3279149 0.3330058 0.3497258 0.3530242 0.3595758
0.3612043 0.3612043 0.4022598
```

#---Looping-----

```
hht=c();
for(i in 1:10000)
{s <- sample(1:N,n,replace=TRUE,prob=pi)
yu=unique(s)
hht[i] <- sum(y[yu]/pii[yu])/N}
```


mean(hht)

Q. Describe the Brewer selection procedure

imp

Answer:

➤ **Brewer's Procedure**

- Brewer (1963a) suggested a selection procedure. The procedure for a sample of size 2 is given as
- Select the first unit with probability proportional to

$$\frac{P_i(1-P_i)}{(1-2P_i)} \quad \text{imp formula}$$

- Select the second unit with probability proportional to size of remaining units.

Q. Describe the Durbin's Procedure

➤ **Durbin's Procedure**

imp mcq

- Durbin (1967) suggested a selection procedure. The procedure for a sample of size 2 is given as
- Select first unit with probability proportional to size.
- Select second unit with probability proportional to size of

$$P_j \left[\frac{1}{1-2P_i} + \frac{1}{1-2P_j} \right] \quad \text{imp formula}$$

Q. Describe Shehbaz-Hanif-Samiuddin's Procedure

- They suggested the procedure in 2003. imp mcqs
- Select first unit with probability proportional to

$$\frac{P_i(1-P_i)}{(1-2P_i)}$$

- Select second unit with probability proportional to size of

$$P_j \left[\frac{1}{1-2P_i} + \frac{1}{1-2P_j} \right] \quad \text{imp formula}$$

Estimation with Auxiliary Variable

MOST IMP MCQ'S

Introduction:

➤ Auxiliary Variable

- We have discussed the estimation of parameters on the basis of single variable.
- Supporting/supplementary variable may be used at design stage or estimation stage.
- Supporting variable is used to enhance the efficiency of estimation.
- The supporting variable must be correlated with main variable.
- The supplementary information usually referred to as benchmark variable or Auxiliary variable.
- Graunt (1662) was the first who used auxiliary information to estimate the population size of England.
- After Graunt(1662), Laplace was the first to introduce the use of auxiliary information for the estimation of population of France.

X = Cell Phones: 2016

Y = Cell Phones: 2017

X = Working hours

Y = Number of Items Processed

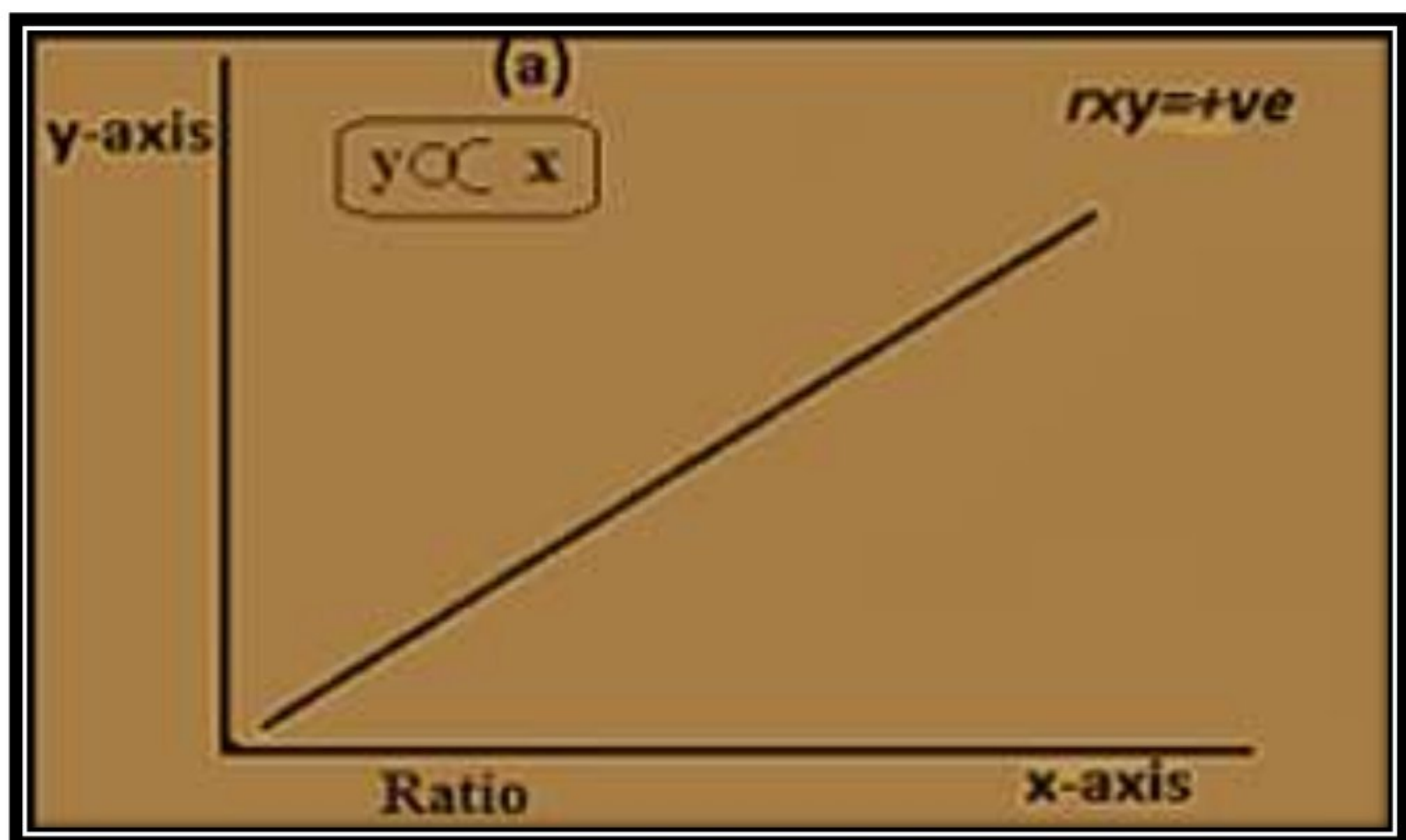
X = the size of unit i

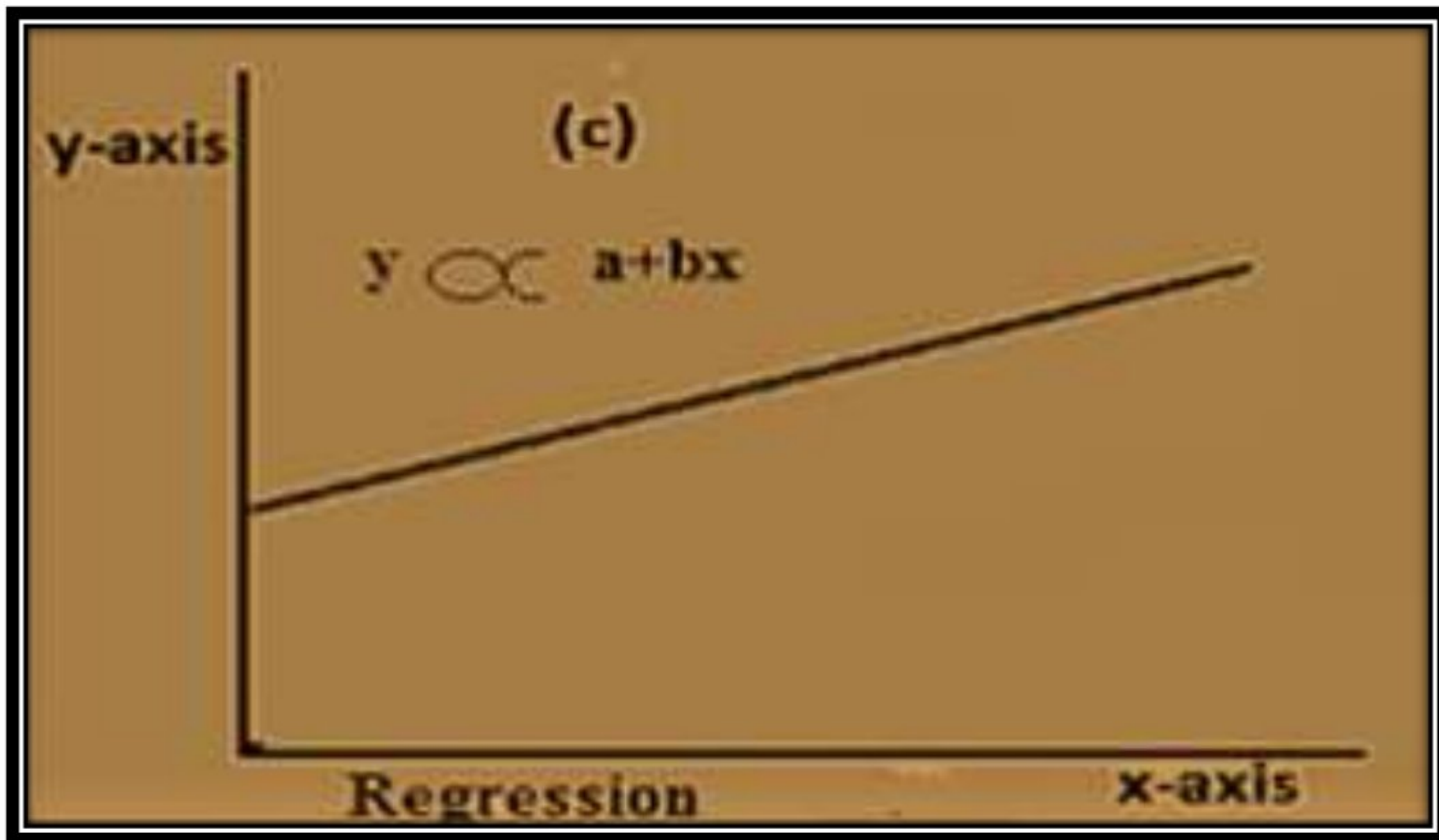
Y = the number of animals in unit i

X = Diameter of the tree

Y = Volume of the tree

➤ Some Types of Relationships





Q. Prove that ratio estimator is almost unbiased for large sample size.

Answer:

➤ **Ratio Estimator**

- The Ratio Estimator is

$$\hat{R} = \frac{y}{x} = \frac{\bar{y}}{\bar{x}}$$

MOST MOST IMP S/Q

➤ **Ratio Estimator for Mean and Total**

- The Ratio Estimator for population mean

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

- The Ratio Estimator for population total

$$y_r = \frac{y}{x} X$$

➤ **Expectation of Ratio Estimator**

- For the large sample size the expectation of ratio estimator is approximately equal to population ratio, i.e.

$$E(\hat{R}) \approx R$$

$$\hat{R} - R = \frac{\bar{y}}{\bar{x}} - R$$

$$= \frac{\bar{y} - R\bar{x}}{\bar{x}}$$

When n is large

$$\hat{R} - R \approx \frac{\bar{y} - R\bar{x}}{\bar{X}}$$

$$E(\hat{R} - R) \approx \frac{E(\bar{y}) - RE(\bar{x})}{\bar{X}}$$

$$E(\hat{R} - R) \approx \frac{E(\bar{y}) - RE(\bar{x})}{\bar{X}}$$

$$E(\hat{R} - R) \approx \frac{\bar{Y} - R\bar{X}}{\bar{X}}$$

$$E(\hat{R}) \approx R$$

Q. Find the variance of Ratio Estimator.

Answer:

➤ **Ratio Estimator**

- The Ratio Estimator is

$$\hat{R} = \frac{y}{x} = \frac{\bar{y}}{\bar{x}}$$

➤ **Ratio Estimator for Mean and Total**

- The Ratio Estimator for population mean

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

- The Ratio Estimator for population total

$$y_r = \frac{y}{x} X$$

➤ **Variance of Ratio Estimator**

$$\hat{R} - R \approx \frac{\bar{y} - R\bar{x}}{\bar{X}}$$

$$E(\hat{R} - R)^2 \approx \frac{1}{\bar{X}^2} E(\bar{y} - R\bar{x})^2$$

$$\bar{d} = \bar{y} - R\bar{x}$$

$$\bar{D} = \bar{Y} - R\bar{X}$$

$$E(\hat{R} - R)^2 \approx \frac{1}{\bar{X}^2} V(\bar{d})$$

MOST IMP L/Q

$$MSE(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2}$$

$$\left[\frac{1}{N-1} \sum_{i=1}^N (Y_i - RX_i)^2 \right]$$

$$mse(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2}$$

$$\left[\frac{1}{n-1} \sum_{i=1}^n (y_i - rx_i)^2 \right]$$

➤ **Alternative Expression of Variance**

$$MSE(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2} \left[\frac{1}{N-1} \sum_{i=1}^N (Y_i - RX_i)^2 \right]$$

$$SE(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2} \frac{1}{N-1} \sum_{i=1}^N [(Y_i - \bar{Y}) - (RX_i - R\bar{X})]^2$$

$$MSE(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2} [S_Y^2 - 2RS_{YX} + R^2 S_X^2]$$

➤ **Expression of Variance for Ratio Estimator of Mean**

$$MSE(\hat{R}) \approx \frac{N-n}{Nn} \frac{1}{\bar{X}^2}$$

$$[S_Y^2 - 2RS_{YX} + R^2 S_X^2]$$

$$MSE(\bar{y}_r) \approx \frac{N-n}{Nn}$$

$$[S_Y^2 - 2R\rho S_Y S_X + R^2 S_X^2]$$

$$MSE(\bar{y}_r) \approx \lambda \bar{Y}^2$$

$$[C_Y^2 - 2\rho C_Y C_X + C_X^2]$$

$$C.L.(\hat{R}) : \hat{R} \pm t \sqrt{mse(\hat{R})}$$

$$C.L.(\bar{y}) : \bar{y}_r \pm t \sqrt{mse(\bar{y}_r)}$$

Q. Find the approximate Bias expression of Ratio Estimator?

Answer:

➤ **Ratio Estimator of Mean**

- The Ratio Estimator is

$$\hat{R} = \frac{y}{x} = \frac{\bar{y}}{\bar{x}}$$

IMP L/Q

➤ **Notations**

$$e_0 = (\bar{y} - \bar{Y}) / \bar{Y}$$

$$e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

Using these notations

$$E(e_0^2) = \lambda C_y^2,$$

$$E(e_1^2) = \lambda C_x^2,$$

$$E(e_0 e_1) = \lambda C_{yx},$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

➤ **Using Notations**

$$\bar{y} = \bar{Y}(1 + e_0)$$

$$\bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

$$\bar{y}_r = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

➤ **Bias of Ratio Estimator**

$$= \bar{Y}(1 + e_0)(1 + e_1)^{-1}$$

$$\approx \bar{Y}(1 + e_0)(1 - e_1 + e_1^2)$$

$$\bar{y}_r - \bar{Y} \approx \bar{Y}(e_0 - e_1 - e_0 e_1 + e_1^2)$$

$$\text{Bias}(\bar{y}_r) \approx \frac{1-f}{n} \bar{Y} (C_x^2 - \rho_{yx} C_y C_x)$$

$$\text{Bias}(\bar{y}_r) \approx \frac{1-f}{n} \bar{Y} (C_x^2 - \rho_{yx} C_y C_x)$$

$$\text{Bias}(\hat{R}) \approx \frac{1-f}{n} R (C_x^2 - \rho_{yx} C_y C_x)$$

Q. Comparison of Ratio Estimator with Mean Per Unit Estimator?

Answer:

➤ **Ratio Estimator for Mean**

- The Ratio Estimator for population mean, i.e.

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

$$\text{Var}(\bar{y}_r) \approx \bar{Y}^2 \lambda$$

$$[C_Y^2 - 2\rho C_Y C_X + C_X^2]$$

- The mean per unit estimator is

$$V(\bar{y}) = \lambda \bar{Y}^2 C_Y^2$$

➤ **Comparison**

S/Q

$$Var(\bar{y}) - Var(\bar{y}_r) > 0$$

$$\lambda \bar{Y}^2 C_Y^2 -$$

$$\lambda \bar{Y}^2 [C_Y^2 - 2\rho C_Y C_X + C_X^2] > 0$$

$$C_Y^2 - [C_Y^2 - 2\rho C_Y C_X + C_X^2] > 0$$

$$2\rho C_Y C_X - C_X^2 > 0$$

$$\rho > \frac{1}{2} \frac{C_X}{C_Y}$$

Q. Prove that Hartley-Ross is an Unbiased Ratio Estimator

$$r_{HR} = \bar{r} + \frac{n(N-1)}{N(n-1)} \frac{(\bar{y} - \bar{r} \bar{x})}{\bar{X}}$$

Answer:

➤ **Hartley-Ross Estimator**

- The HR Estimator is

$$r_{HR} = \bar{r} + \frac{n(N-1)}{N(n-1)} \frac{(\bar{y} - \bar{r} \bar{x})}{\bar{X}}$$

$$\text{where, } \bar{r} = \frac{1}{n} \sum_{i=1}^n \frac{y_i}{x_i}$$

- The variance of HR Estimator is

$$Var(r_{HR}) \approx \frac{1}{n \bar{X}^2}$$

$$[S_Y^2 + \bar{R}^2 S_X^2 - 2\bar{R} P S_Y S_X]$$

- We know that

$$E(\bar{r}) = E\left(\sum_{i=1}^n \frac{r_i}{n}\right) = E(r_i)$$

$$E(\bar{r}) = \frac{E(r_i)E(x_i)}{E(x_i)}$$

$$= \frac{E(r_i)E(x_i)}{\bar{X}}$$

$$E(\bar{r}) - R = \frac{E(r_i)E(x_i)}{\bar{X}} - R$$

$$E(\bar{r}) - R =$$

$$-\frac{1}{\bar{X}} [\bar{Y} - E(r_i)E(x_i)]$$

$$\begin{aligned}
 E(\bar{r}) - R &= \\
 -\frac{1}{\bar{X}} [\bar{Y} - E(r_i)E(x_i)] \\
 E(\bar{r}) - R &= \\
 -\frac{1}{\bar{X}} [E(r_i x_i) - E(r_i)E(x_i)] \\
 \bar{Y} &= E(y_i) = E\left(\frac{y_i}{x_i} x_i\right) = E(r_i x_i) \\
 E(\bar{r}) - R &= -\frac{1}{\bar{X}} \text{cov}(r_i x_i) \\
 E(\bar{r}) - R &= \\
 -\frac{1}{\bar{X}} \frac{1}{N} \sum_{i=1}^N (r_i - R)(x_i - \bar{X}) \\
 E(\bar{r}) - R &= -\frac{1}{\bar{X}} \frac{N-1}{N} E(s_{rx}) \\
 E(\bar{r}) - R &= \\
 -\frac{1}{\bar{X}} \frac{N-1}{N(N-1)} \sum_{i=1}^N (r_i - R)(x_i - \bar{X}) \\
 E(\bar{r}) - R &= -\frac{1}{\bar{X}} \frac{N-1}{N} E(s_{rx}) \\
 E(\bar{r}) - R &= \\
 -\frac{1}{\bar{X}} \frac{N-1}{N(N-1)} \sum_{i=1}^N (r_i - R)(x_i - \bar{X})
 \end{aligned}$$

➤ **Hartley-Ross Estimator**

$$\begin{aligned}
 s_{rx} &= \frac{1}{n-1} \left(\sum_{i=1}^n r_i x_i - \frac{\sum_{i=1}^n r_i \sum_{i=1}^n x_i}{n} \right) = \frac{1}{n-1} \left(\sum_{i=1}^n r_i x_i - n \bar{r} \bar{x} \right) \\
 s_{rx} &= \frac{1}{n-1} \left(\sum_{i=1}^n y_i - n \bar{x} \bar{r} \right) = \frac{n}{n-1} (\bar{y} - \bar{r} \bar{x}) \\
 E(\bar{r}) - R &= -\frac{1}{\bar{X}} \frac{N-1}{N} E(s_{rx}) = -\frac{1}{\bar{X}} \frac{N-1}{N} E\left(\frac{n}{n-1} \bar{y} - \bar{r} \bar{x}\right) \\
 E(\bar{r}) + E\left[\frac{1}{\bar{X}} \frac{N-1}{N} \frac{n}{n-1} (\bar{y} - \bar{r} \bar{x})\right] &= R \\
 E(\bar{r}) + E\left[\frac{1}{\bar{X}} \frac{N-1}{N} \frac{n}{n-1} (\bar{y} - \bar{r} \bar{x})\right] &= R
 \end{aligned}$$

$$E\left[\bar{r} + \frac{1}{\bar{X}} \frac{N-1}{N} \frac{n}{n-1} (\bar{y} - \bar{r}\bar{x})\right] = R$$

➤ **HR Unbiased Ratio Estimator**

$$E\left[\bar{r} + \frac{1}{\bar{X}} \frac{N-1}{N} \frac{n}{n-1} (\bar{y} - \bar{r}\bar{x})\right] = R$$

$$E[\text{Estimator}] = \text{Parameter}$$

$$r_{HR} = \bar{r} + \frac{n(N-1)}{N(n-1)} \frac{(\bar{y} - \bar{r}\bar{x})}{\bar{X}}$$

Q. Prove that regression estimator is an unbiased estimator.

Ans:

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx} (\bar{X} - \bar{x})$$

$$E(\bar{y}_{reg}) = E(\bar{y}) + \hat{\beta}_{yx} E(\bar{X} - \bar{x})$$

$$E(\bar{y}_{reg}) = \bar{Y}$$

IMP S/Q

Q. Find variance of Regression Estimator.

Ans

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx} (\bar{X} - \bar{x})$$

$$\bar{y}_{reg} = \bar{Y}(1 + e_0) +$$

$$\hat{\beta}_{yx} (\bar{X} - \bar{X}(1 + e_1))$$

$$\bar{y}_{reg} = \bar{Y}(1 + e_0) - \hat{\beta}_{yx} \bar{X}e_1$$

$$\bar{y}_{reg} = \bar{Y}(1 + e_0) - \hat{\beta}_{yx} \bar{X}e_1$$

$$\bar{y}_{reg} - \bar{Y} = \bar{Y}e_0 - \hat{\beta}_{yx} \bar{X}e_1$$

$$E(\bar{y}_{reg} - \bar{Y})^2 = E(\bar{Y}e_0 - \hat{\beta}_{yx} \bar{X}e_1)^2$$

$$V(\bar{y}_{reg}) = E(\bar{Y}e_0)^2 + E(\hat{\beta}_{yx} \bar{X}e_1)^2$$

$$-2E(\bar{Y}e_0 \hat{\beta}_{yx} \bar{X}e_1)$$

$$= E(\bar{Y}e_0)^2 + E(\hat{\beta}_{yx} \bar{X}e_1)^2$$

$$-2E(\bar{Y}e_0 \hat{\beta}_{yx} \bar{X}e_1)$$

$$= \lambda \left[\bar{Y}^2 C_y^2 + \hat{\beta}_{yx}^2 \bar{X}^2 C_x^2 - 2\bar{Y}\bar{X} \hat{\beta}_{yx} \rho C_y C_x \right]$$

S/Q

$$\begin{aligned}
 &= \lambda \begin{bmatrix} \bar{Y}^2 C_y^2 + \rho^2 \frac{S_y^2}{S_x^2} \bar{X}^2 C_x^2 \\ -2\bar{Y}\bar{X} \rho \frac{S_y}{S_x} \rho C_y C_x \end{bmatrix} \\
 &= \lambda \begin{bmatrix} \bar{Y}^2 C_y^2 + \rho^2 \frac{S_y^2}{S_x^2} \bar{X}^2 C_x^2 \\ -2\bar{Y}\bar{X} \rho \frac{S_y}{S_x} \rho C_y C_x \end{bmatrix} = \lambda \begin{bmatrix} \bar{Y}^2 C_y^2 + \rho^2 \frac{S_y^2}{S_x^2} \bar{X}^2 C_x^2 \\ -2\bar{Y}\bar{X} \rho \frac{S_y}{S_x} \rho C_y C_x \end{bmatrix} \\
 &V(\bar{y}_{reg}) = \lambda S_y^2 [1 - \rho^2]
 \end{aligned}$$

Q. Find expected value and MSE of ratio estimator by taking all possible samples of size 2 from the following population.

Unit No.	1	2	3	4
Y_i	5	12	15	28
X_i	11	21	32	14

Answer:

➤ **Small Population Example**

- The following is an artificial population with four values.

Unit No.	1	2	3	4
Y_i	5	12	15	28
X_i	11	21	32	14

- We are interested to take all possible samples of size 2.
- Calculate expected value for the ratio estimator.

Unit No.	1	2	3	4
Y_i	5	12	15	28
X_i	11	21	32	14

$$\begin{aligned}
 &\frac{\bar{y}}{\bar{x}} \\
 &\bar{X} = 19.5
 \end{aligned}$$

IMP L/Q

Unit	$\bar{y}$	$\bar{x}$	$\bar{y} / \bar{x}$	$\bar{y}_r$
1 , 2	8.5	16	0.53	10.36
1 , 3	10	21.5	0.47	9.07
1 , 4	16.5	12.5	1.32	25.74
2 , 3	13.5	26.5	0.51	9.93
2 , 4	20	17.5	1.14	22.29
3 , 4	21.5	23	0.93	18.23

$$MSE(\bar{y}_r) = \sum_{i=1}^6 (\bar{y}_r - 15)^2 \frac{1}{6}$$

$$MSE(\bar{y}_r) = 43.54$$

$$MSE(\bar{y}_r) = \sum_{i=1}^6 (\bar{y}_{ri} - \bar{Y})^2 P(s)$$

Q. Read the data from the following command and select 10000 samples by simple random smapling without replacement. Find ratio estimator from each sample. Further find mean of ratio estimators and compare it with mean per unit estimator.

```
y <- trees$Volume
zi <- trees$Girth
```

Answer:

➤ Example

- The Ratio Estimator is

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

➤ Defining Population

```
#----Defining Population
y <- trees$Volume
x <- trees$Girth
N <- 31; n <- 4
mux <- mean(x)
r<-c();mratio<-c();
```

IMP S/Q

➤ Relative Efficiency


```
#----repeating 10000 times
for(i in 1:10000)
{ s<-sample(1:N,n)
m[i]<-mean(y[s])
r[i] <- mean(y[s])/mean(x[s])
mratio[i] <- r[i]*mux }
var(m);var(mratio)
re<-var(m)/var(mratio);re
```

Q. Generate a population of size 1000 for the given parameters using bivariate normal distribution, such that

$$[Y, X] \text{ as } \mu = \begin{bmatrix} 2 & 2 \end{bmatrix}.$$

$$\Sigma = \begin{bmatrix} 1 & 0.85 \\ 0.85 & 1 \end{bmatrix}$$

We consider sample sizes: $n = 10$. Select 10,000 random samples considering SRSWOR and calculate

$$t = \bar{y} \frac{\bar{X}}{\bar{x}}$$

Further, calculate the MSEs for above estimators and calculate relative efficiencies with respect to mean per unit estimator ($\bar{y}$)

Answer:

```
library(mvtnorm)
N=1000;ryx=0.85; n=10;
m=c(2,2); # vector of mean.
# variance covariance matrix is given below.
sig=matrix(c(1,0.85,0.85,1),ncol=2);
r=rmvnorm(N,m,sig);
x=r[,2];y=r[,1];
data=data.frame(x,y);
plot(y,x)
```

S/Q

➤ Simulation Study

```
r<-c();mratio<-c();m=c();
for(i in 1:10000)
{ s<-sample(1:N,n)
m[i]<-mean(y[s])
r[i] <- mean(y[s])/mean(x[s])
mratio[i] <- r[i]*mux }
var(m);var(mratio)
re<-var(m)/var(mratio);re
```


Q. Find Bias of Product Estimator?**Answer:****➤ Product Estimator**

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

$$\bar{y}_p = \frac{\bar{y}\bar{x}}{\bar{X}}$$

➤ Bias of Product Estimator

$$e_0 = (\bar{y} - \bar{Y}) / \bar{Y}$$

$$e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

Using these notations,

$$E(e_0^2) = \lambda C_y^2,$$

$$E(e_1^2) = \lambda C_x^2,$$

$$E(e_0 e_1) = \lambda C_{yx},$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

➤ Using Notations

$$\bar{y} = \bar{Y}(1 + e_0)$$

$$\bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_p = \frac{\bar{y}\bar{x}}{\bar{X}}$$

$$\bar{y}_p = \frac{\bar{Y}(1 + e_0)\bar{X}(1 + e_1)}{\bar{X}}$$

$$= \bar{Y}(1 + e_0)(1 + e_1)$$

➤ Bias of Product Estimator

$$= \bar{Y}(1 + e_0)(1 + e_1)$$

$$= \bar{Y}(1 + e_0 + e_1 + e_0 e_1)$$

$$\bar{y}_p - \bar{Y} = \bar{Y}(e_0 + e_1 + e_0 e_1)$$

$$E(\bar{y}_p - \bar{Y}) = \bar{Y}E(e_0 + e_1 + e_0 e_1)$$

$$\text{Bias}(\bar{y}_p) = \bar{Y}E(e_0 + e_1 + e_0 e_1)$$

$$\text{Bias}(\bar{y}_p) = \bar{Y}E(e_0 e_1)$$

$$E(e_0 e_1) = \lambda C_{yx},$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

V.V. IMP LONG

PAST PAPER SOMETIME
BIASE SOMETIME MSE

Regard: Jannat Rana

$$\text{Bias}(\bar{y}_p) = \bar{Y} \lambda C_{yx}$$

Q. Find Mean Square Error of Product Estimator?

Answer:

➤ **MSE of Product Estimator**

$$\bar{y}_p = \frac{\bar{y}\bar{x}}{\bar{X}}$$

$$e_0 = (\bar{y} - \bar{Y}) / \bar{Y}$$

$$e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

Using these notations,

$$E(e_0^2) = \lambda C_y^2,$$

$$E(e_1^2) = \lambda C_x^2,$$

$$E(e_0 e_1) = \lambda C_{yx},$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

➤ **Using Notations**

$$\bar{y}_p = \frac{\bar{y}\bar{x}}{\bar{X}}$$

$$\bar{y}_p = \frac{\bar{Y}(1+e_0)\bar{X}(1+e_1)}{\bar{X}}$$

$$= \bar{Y}(1+e_0)(1+e_1)$$

➤ **MSE of Product Estimator**

$$= \bar{Y}(1+e_0)(1+e_1)$$

$$= \bar{Y}(1+e_0+e_1+e_0e_1)$$

$$\bar{y}_p - \bar{Y} = \bar{Y}(e_0+e_1+e_0e_1)$$

$$E(\bar{y}_p - \bar{Y}) = \bar{Y}E(e_0+e_1+e_0e_1)$$

$$E(\bar{y}_p - \bar{Y})^2 = \bar{Y}^2 E(e_0+e_1+e_0e_1)^2$$

$$MSE(\bar{y}_p) = \bar{Y}^2 E(e_0+e_1+e_0e_1)^2$$

$$MSE(\bar{y}_p) \approx \bar{Y}^2 E(e_0+e_1)^2$$

$$MSE(\bar{y}_p) \approx \bar{Y}^2 E(e_0^2 + e_1^2 + 2e_0e_1)$$

$$MSE(\bar{y}_p) \approx$$

$$\lambda \bar{Y}^2 (C_y^2 + C_x^2 + 2\rho C_y C_x)$$

$$MSE(\bar{y}_p) \approx$$

$$\lambda \bar{Y}^2 (C_y^2 + C_x^2 - 2\rho C_y C_x)$$

MOST IMP LONG

Q. Find expected value and MSE of product estimator by taking all possible samples of size 2 from the following population.

Unit No.	1	2	3	4
Y_i	5	12	15	28
X_i	32	21	14	11

Answer:

➤ **Small Population Example**

IMP L/Q

➤ The following is an artificial population with four values.

Unit No.	1	2	3	4
Y_i	5	12	15	28
X_i	32	21	14	11

➤ We are interested to take all possible samples of size 2.

➤ Calculate expected value for the product estimator.

$$\frac{\overline{yx}}{\bar{X}}$$

$$\bar{X} = 19.5$$

➤ **All Possible Samples**

Unit	1	2	3	4
No.				
Y_i	5	12	15	28
X_i	32	21	14	11

Unit	$\bar{y}$	$\bar{x}$	$\bar{y} \times \bar{x}$	$\bar{y}_p$
1, 2	8.5	26.5	225.25	11.55
1, 3	10	23	230	11.79
1, 4	16.5	21.5	354.75	18.19
2, 3	13.5	17.5	236.25	12.12
2, 4	20	16	320	16.41
3, 4	21.5	12.5	268.75	13.78

➤ Expected Value

Unit			
1, 2	8.5	26.5	11.55
1, 3	10	23	11.79
1, 4	16.5	21.5	18.19
2, 3	13.5	17.5	12.12
2, 4	20	16	16.41
3, 4	21.5	12.5	13.78

REPEAT EXAM

$$popmean = 15 \quad E(\bar{y}_p) = 13.97$$

$$MSE(\bar{y}_p) = \sum_{i=1}^6 (\bar{y}_{pi} - \bar{Y})^2 P(s)$$

$$MSE(\bar{y}_r) = \sum_{i=1}^6 (\bar{y}_r - 15)^2 \frac{1}{6}$$

$$MSE(\bar{y}_p) = 7.36$$

Q.109 Generate a population of size 1000 for the given parameters using bivariate normal distribution, such that

$$[Y, X] \text{ as } \mu = [2 \ 2].$$

IMP S/Q

$$\Sigma = \begin{bmatrix} 1 & -0.85 \\ -0.85 & 1 \end{bmatrix}$$

We consider sample sizes: $n = 10$. Select 10,000 random samples considering SRSWOR and calculate

$$t = \bar{y} \frac{\bar{x}}{\bar{X}}$$

Further calculate the MSEs for above estimators and calculate relative efficiencies with respect to mean per unit estimator ($\bar{y}$)

Answer:

➤ **Defining Population**

```
library(mvtnorm)
N=1000;ryx=-0.85; n=10;
m=c(2,2); # vector of mean.
# variance covariance matrix is given below.
sig=matrix(c(1,-0.85,-0.85,1),ncol=2);
r=rmvnorm(N,m,sig);
x=r[,2];y=r[,1];
data=data.frame(x,y);
```

IMP MCQ'S

```
p<-c();mp<-c();m=c();
for(i in 1:10000)
{ s<-sample(1:N,n)
m[i]<-mean(y[s])
p[i] <- mean(y[s])*mean(x[s])
mp[i] <- p[i]/mux }
var(m);var(mp)
re<-var(m)/var(mp);re
```


Ratio Estimator in Stratified Sampling

Q. Find the expression of Bias of Combined Type Ratio Estimator.

Answer:

➤ **Combined Type Ratio Estimator in Stratified Sampling**

- The combined type ratio estimator is

$$\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X} \quad \text{most imp mcq's}$$

➤ **Notations**

$$e_{0st} = (\bar{y}_{st} - \bar{Y}) / \bar{Y}$$

$$e_{1st} = (\bar{x}_{st} - \bar{X}) / \bar{X}$$

Using these notations,

$$E(e_{0st}^2) = \sum_{h=1}^k W_h^2 \lambda_h C_{yh}^2,$$

$$E(e_{1st}^2) = \sum_{h=1}^k W_h^2 \lambda_h C_{xh}^2,$$

$$E(e_{0st} e_{1st}) = \sum_{h=1}^k W_h^2 \lambda_h C_{yxh}$$

$$\text{where, } C_{yxh} = \rho_{yxh} C_{yh} C_{xh}$$

short question

➤ **Bias of the Estimator**

$$\bar{y}_{st} = \bar{Y} (1 + e_{0st})$$

$$\bar{x}_{st} = \bar{X} (1 + e_{1st})$$

$$\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X}$$

$$\bar{y}_{rst} = \frac{\bar{Y} (1 + e_{0st})}{\bar{X} (1 + e_{1st})} \bar{X}$$

➤ **Bias of Ratio Estimator**

$$= \bar{Y} (1 + e_{0st}) (1 + e_{1st})^{-1}$$

$$\approx \bar{Y} (1 + e_{0st}) (1 - e_{1st} + e_{1st}^2)$$

$$\bar{y}_{rst} - \bar{Y} \approx \bar{Y} \begin{pmatrix} e_{0st} - e_{1st} \\ -e_{0st} e_{1st} + e_{1st}^2 \end{pmatrix}$$

$$E(e_{0st} e_{1st}) = \sum_{h=1}^k W_h^2 \lambda_h C_{yxh}$$

$$\text{Bias}(\bar{y}_{rst}) \approx$$

$$\bar{Y} \sum_{h=1}^k W_h^2 \lambda_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

Q. Find the expression of MSE of Combined Type Ratio Estimator?

Answer:

➤ **Combined Ratio Estimator in Stratified Sampling**

- The combined type ratio estimator is

$$\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X}$$

➤ **Notations**

$$e_{0st} = (\bar{y}_{st} - \bar{Y}) / \bar{Y}$$

$$e_{1st} = (\bar{x}_{st} - \bar{X}) / \bar{X}$$

Using these notations

$$E(e_{0st}^2) = \sum_{h=1}^k W_h^2 \lambda_h C_{yh}^2,$$

$$E(e_{1st}^2) = \sum_{h=1}^k W_h^2 \lambda_h C_{xh}^2,$$

$$E(e_{0st} e_{1st}) = \sum_{h=1}^k W_h^2 \lambda_h C_{yxh}$$

$$\text{where, } C_{yxh} = \rho_{yxh} C_{yh} C_{xh}$$

notation

most most imp mcq's

➤ **MSE of Combined Type Ratio Estimator**

$$\bar{y}_{st} = \bar{Y}(1 + e_{0st})$$

$$\bar{x}_{st} = \bar{X}(1 + e_{1st})$$

$$\bar{y}_{rst} = \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X}$$

$$\bar{y}_{rst} = \frac{\bar{Y}(1 + e_{0st})}{\bar{X}(1 + e_{1st})} \bar{X}$$

$$= \bar{Y} (1 + e_{0st}) (1 + e_{1st})^{-1}$$

$$\approx \bar{Y} (1 + e_{0st}) (1 - e_{1st})$$

$$\bar{y}_{rst} \approx \bar{Y} (1 + e_{0st} - e_{1st})$$

$$\bar{y}_{rst} - \bar{Y} \approx \bar{Y} (e_{0st} - e_{1st})$$

$$E(\bar{y}_{rst} - \bar{Y})^2 \approx \bar{Y}^2 E(e_{0st} - e_{1st})^2$$

prove long question

feb2024

most imp

$$\begin{aligned}\bar{y}_{rst} &\approx \bar{Y}(1 + e_{0st} - e_{1st} - e_{0st}e_{1st}) \\ E(\bar{y}_{rst} - \bar{Y})^2 &\approx \bar{Y}^2 E(e_{0st} - e_{1st})^2 \\ E(\bar{y}_{rst} - \bar{Y})^2 &\approx \\ \bar{Y}^2 &\left(E(e_{0st}^2) + E(e_{1st}^2) \right. \\ &\quad \left. - 2E(e_{1st}e_{0st}) \right) \\ MSE(\bar{y}_{rst}) &\approx \\ \bar{Y}^2 \sum_{h=1}^L W_h^2 \gamma_h &\begin{bmatrix} C_{yh}^2 + C_{xh}^2 \\ -2C_{yxh} \end{bmatrix} \\ MSE(\bar{y}_{rst}) &\approx \\ \bar{Y}^2 \sum_{h=1}^L W_h^2 \gamma_h &\begin{bmatrix} C_{yh}^2 + C_{xh}^2 \\ -2C_{yxh} \end{bmatrix}\end{aligned}$$

➤ **Example**

- Consider a population of size 700 consisting on three strata such that $N_1=100$, $N_2=250$ and $N_3=350$. The required sample size is 18.
- The population consisting on two variables Y and X.
- The population mean for the Variable Y and X is 15 and 62.14, respectively.
- The sample size from stratum-1, stratum-2 and stratum-3 is arbitrarily decided as 4, 8 and 6, respectively.
- The sample from each stratum is chosen as (y,x)

Stra-1	Stra-2	Stra-3
1,22	7,39	23,92
3,29	12,55	14,65
2,25	8,42	20,84
5,32	5,30	22,89
	11,51	24,96

10,49 17,68

9,45

12,54

➤ Considering the Variable Y

	Stra-1	Stra-2	Stra-3
	1	7	23
	3	12	14
	2	8	20
	5	5	22
		11	24
		10	17
		9	
		12	
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

➤ Considering the Variable X

	Stra-1	Stra-2	Stra-3
	22	39	92
	29	55	65
	25	42	84
	32	30	89
		51	96
		49	68
		45	
		54	
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91

n_h	4	8	6
-------	---	---	---

➤ **Sample Means**

- Sample mean of variable Y

$$\bar{y}_{st} = \sum_{h=1}^k W_h \bar{y}_h = \sum_{h=1}^k N_h \bar{y}_h / N$$

$$\bar{y}_{st} = \frac{1}{N} [N_1 \bar{y}_1 + N_2 \bar{y}_2 + N_3 \bar{y}_3]$$

- Sample mean of variable X

$$\bar{x}_{st} = \sum_{h=1}^k W_h \bar{x}_h = \sum_{h=1}^k N_h \bar{x}_h / N$$

$$\bar{x}_{st} = \frac{1}{N} [N_1 \bar{x}_1 + N_2 \bar{x}_2 + N_3 \bar{x}_3]$$

➤ **Sample mean of Y**

	Stra-1	2	Stra-3
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

$$\bar{y}_{st} = \sum_{h=1}^k W_h \bar{y}_h = \sum_{h=1}^k N_h \bar{y}_h / N$$

$$\bar{y}_{st} = \frac{1}{N} [N_1 \bar{y}_1 + N_2 \bar{y}_2 + N_3 \bar{y}_3]$$

$$\bar{y}_{st} = 13.70$$

➤ **Sample mean of X**

	Stra-1	Stra-2	Stra-3
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91
n_h	4	8	6

$$\bar{x}_{st} = \sum_{h=1}^k W_h \bar{x}_h = \sum_{h=1}^k N_h \bar{x}_h / N$$

$$\bar{x}_{st} = \frac{1}{N} [N_1 \bar{x}_1 + N_2 \bar{x}_2 + N_3 \bar{x}_3]$$

$$\bar{x}_{st} = 61.32$$

➤ **Combined Ratio Estimator**

- The combined type ratio estimator is

$$\begin{aligned}\bar{y}_{rst} &= \frac{\bar{y}_{st}}{\bar{x}_{st}} \bar{X} \\ \bar{y}_{rst} &= \frac{13.70}{61.32} \times 62.14 \\ \bar{y}_{rst} &= 13.89\end{aligned}$$

mcq's

Q.115 Find the expression of Bias and MSE of Separate Type Ratio Estimator in stratified sampling.

Answer:

➤ **Separate Type Ratio Estimator**

- The separate type ratio estimator is

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^k W_h \bar{y}_{rh} \\ \bar{y}_{rsst} &= \sum_{h=1}^k W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h\end{aligned}$$

➤ **Bias of Ratio Estimator**

$$Bias(\bar{y}_r) \approx$$

$$\frac{1-f}{n} \bar{Y} (C_x^2 - \rho_{yx} C_y C_x)$$

- In case of Stratified Sampling

$$Bias(\bar{y}_{rh}) \approx$$

$$\frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

$$E(\bar{y}_{rh}) - \bar{Y}_h \approx$$

$$\frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

$$E(\bar{y}_{rh}) \approx \bar{Y}_h +$$

$$\frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

➤ **Bias of Separate Type Ratio Estimator**

$$\bar{y}_{rsst} = \sum_{h=1}^k W_h \bar{y}_{rh} \Rightarrow E(\bar{y}_{rsst}) = \sum_{h=1}^k W_h E(\bar{y}_{rh})$$

$$E(\bar{y}_{rsst}) = \sum_{h=1}^k W_h \left(\bar{Y}_h + \frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh}) \right)$$

$$E(\bar{y}_{rsst}) = \bar{Y} + \sum_{h=1}^k W_h \frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

$$E(\bar{y}_{rsst}) - \bar{Y} = \sum_{h=1}^k W_h \frac{1-f_h}{n_h} \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

➤ **Bias of Separate Type Ratio Estimator**

$$\text{Bias}(\bar{y}_{rsst}) = \sum_{h=1}^k W_h \lambda_h \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

short

➤ **Expression of MSE for Ratio Estimator of Mean**

$$MSE(\bar{y}_r) \approx$$

$$\lambda \bar{Y}^2 (C_y^2 + C_x^2 - 2\rho_{yx} C_y C_x)$$

$$MSE(\bar{y}_{rh}) \approx$$

$$\lambda_h \bar{Y}_h^2 (C_{yh}^2 + C_{xh}^2 - 2\rho_{yxh} C_{yh} C_{xh})$$

➤ **MSE of Separate Type Ratio Estimator**

$$\bar{y}_{rsst} = \sum_{h=1}^k W_h \bar{y}_{rh} \Rightarrow MSE(\bar{y}_{rsst}) = \sum_{h=1}^k W_h^2 MSE(\bar{y}_{rh})$$

$$MSE(\bar{y}_{rh}) \approx \lambda_h \bar{Y}_h^2 (C_{yh}^2 + C_{xh}^2 - 2\rho_{yxh} C_{yh} C_{xh})$$

$$MSE(\bar{y}_{rsst}) \approx \sum_{h=1}^k W_h^2 \lambda_h \bar{Y}_h^2 (C_{yh}^2 + C_{xh}^2 - 2\rho_{yxh} C_{yh} C_{xh})$$

$$MSE(\bar{y}_{rsst}) \approx \sum_{h=1}^k \left[W_h^2 \lambda_h \frac{\sum_{i=1}^{N_h} (Y_{hi} - R_h X_{hi})^2}{N_h - 1} \right]$$

➤ **Bias and MSE of Separate Type Ratio Estimator**

$$\text{Bias}(\bar{y}_{rsst}) =$$

$$\sum_{h=1}^k W_h \lambda_h \bar{Y}_h (C_{xh}^2 - \rho_{yxh} C_{yh} C_{xh})$$

$$MSE(\bar{y}_{rsst}) \approx$$

$$\sum_{h=1}^k W_h^2 \lambda_h \bar{Y}_h^2 \begin{pmatrix} C_{yh}^2 + C_{xh}^2 \\ -2\rho_{yxh} C_{yh} C_{xh} \end{pmatrix}$$

Example

- Consider a population of size 700 consisting on three strata such that $N_1=100$, $N_2=250$ and $N_3=350$. The required sample size is 18.
- The population consisting on two variables Y and X.
- The population means for the Variable Y and X is 15 and 62.14, respectively.
- The sample size from stratum-1, stratum-2 and stratum-3 is arbitrarily decided as 4, 8 and 6, respectively.
- The overall mean of stratum-1, stratum-2 and stratum-3 for variable X is 25, 45 and 85, respectively.
- The sample from each stratum is chosen as (y,x)

Stra-1	Stra-2	Stra-3
1,22	7,39	23,92
3,29	12,55	14,65
2,25	8,42	20,84
5,32	5,30	22,89
	11,51	24,96
	10,49	17,68
	9,45	
	12,54	

➤ **Considering the Variable Y**

	Stra-1	Stra-2	Stra-3
	1	7	23
	3	12	14
	2	8	20
	5	5	22
		11	24
		10	17
		9	
		12	
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

➤ Considering the Variable X

	Stra-1	Stra-2	Stra-3
	22	39	92
	29	55	65
	25	42	84
	32	30	89
		51	96
		49	68
		45	
		54	
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91
n_h	4	8	6

➤ Sample Means

- Sample mean of variable Y

$$\bar{y}_{st} = \sum_{h=1}^k W_h \bar{y}_h = \sum_{h=1}^k N_h \bar{y}_h / N$$

$$\bar{y}_{st} = \frac{1}{N} [N_1 \bar{y}_1 + N_2 \bar{y}_2 + N_3 \bar{y}_3]$$

- Sample mean of variable X

$$\bar{x}_{st} = \sum_{h=1}^k W_h \bar{x}_h = \sum_{h=1}^k N_h \bar{x}_h / N$$

$$\bar{x}_{st} = \frac{1}{N} [N_1 \bar{x}_1 + N_2 \bar{x}_2 + N_3 \bar{x}_3]$$

➤ Sample Information of Y and X

- Sample mean of variable Y

	Stra-1	Stra-2	Stra-3
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

- Sample mean of variable X

	Stra-1	Stra-2	Stra-3
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91
n_h	4	8	6

$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h$$

Sample mean of variable

$$= W_1 \frac{\bar{y}_1}{\bar{x}_1} \bar{X}_1 + W_2 \frac{\bar{y}_2}{\bar{x}_2} \bar{X}_2 + W_3 \frac{\bar{y}_3}{\bar{x}_3} \bar{X}_3$$

$$= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}$$

$$\bar{y}_{r1} = \frac{\bar{y}_1}{\bar{x}_1} \bar{X}_1 \quad \text{imp formula}$$

➤ Separate Type Ratio Estimator

$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h$$

$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \bar{y}_{rh}$$

$$= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}$$

$N_1=100$,
 $N_2=250$ and $N_3=350$

$W_1=0.142857$	Means
$W_2=0.357143$	2.546296
$W_3=0.5$	9.122288
	20.64861

➤ Separate Type Ratio Estimator

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h \\ \bar{y}_{rsst} &= \sum_{h=1}^3 W_h \bar{y}_{rh} \\ &= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}\end{aligned}$$

$$W_1 = 0.142857$$

$$W_2 = 0.357143$$

$$W_3 = 0.5$$

➤ **Separate Type Ratio Estimator**

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h \\ \bar{y}_{rsst} &= \sum_{h=1}^3 W_h \bar{y}_{rh} \\ &= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}\end{aligned}$$

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$$\begin{aligned}&= 0.143 \times 2.55 + 0.357 \times 9.12 \\ &\quad + 0.5 \times 20.65 \\ &= 13.95\end{aligned}$$

Example

- Consider a population of size 700 consisting on three strata such that $N_1=100$, $N_2=250$ and $N_3=350$. The required sample size is 18.
- The population consisting on two variables Y and X.
- The population means for the Variable Y and X is 15 and 62.14, respectively.
- The sample size from stratum-1, stratum-2 and stratum-3 is arbitrarily decided as 4, 8 and 6, respectively.
- The overall mean of stratum-1, stratum-2 and stratum-3 for variable X is 25, 45 and 85, respectively.
- The sample from each stratum is chosen as (y,x)

Stra-1	Stra-2	Stra-3
1,22	7,39	23,92
3,29	12,55	14,65
2,25	8,42	20,84
5,32	5,30	22,89
	11,51	24,96
	10,49	17,68
	9,45	
	12,54	

➤ Considering the Variable Y

	Stra-1	Stra-2	Stra-3
	1	7	23
	3	12	14
	2	8	20
	5	5	22
		11	24
		10	17
		9	
		12	
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

➤ Considering the Variable X

	Stra-1	Stra-2	Stra-3
	22	39	92
	29	55	65
	25	42	84
	32	30	89
		51	96
		49	68
		45	
		54	
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91
n_h	4	8	6

➤ **Sample Means**

- Sample mean of variable Y

$$\bar{y}_{st} = \sum_{h=1}^k W_h \bar{y}_h = \sum_{h=1}^k N_h \bar{y}_h / N$$

$$\bar{y}_{st} = \frac{1}{N} [N_1 \bar{y}_1 + N_2 \bar{y}_2 + N_3 \bar{y}_3]$$

- Sample mean of variable X

$$\bar{x}_{st} = \sum_{h=1}^k W_h \bar{x}_h = \sum_{h=1}^k N_h \bar{x}_h / N$$

$$\bar{x}_{st} = \frac{1}{N} [N_1 \bar{x}_1 + N_2 \bar{x}_2 + N_3 \bar{x}_3]$$

➤ **Sample Information of Y and X**

- Sample mean of variable Y

	Stra-1	Stra-2	Stra-3
mean	2.75	9.25	20
N_h	100	250	350
S_h	1.708	2.493	3.847
n_h	4	8	6

- Sample mean of variable X

	Stra-1	Stra-2	Stra-3
mean	27	45.63	82.33
N_h	100	250	350
S_h	4.397	8.450	12.91
n_h	4	8	6

$$\bar{y}_{rsst} = \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h$$

$$= W_1 \frac{\bar{y}_1}{\bar{x}_1} \bar{X}_1 + W_2 \frac{\bar{y}_2}{\bar{x}_2} \bar{X}_2 + W_3 \frac{\bar{y}_3}{\bar{x}_3} \bar{X}_3$$

$$= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}$$

$$\bar{y}_{r1} = \frac{\bar{y}_1}{\bar{x}_1} \bar{X}_1$$

➤ **Separate Type Ratio Estimator**

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h \\ \bar{y}_{rsst} &= \sum_{h=1}^3 W_h \bar{y}_{rh} \\ &= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}\end{aligned}$$

$$W_1=0.142857$$

$$W_2=0.357143$$

$$W_3= 0.5$$

➤ **Separate Type Ratio Estimator**

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h \\ \bar{y}_{rsst} &= \sum_{h=1}^3 W_h \bar{y}_{rh} \\ &= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3}\end{aligned}$$

Means
2.546296
9.122288
20.64861

$$W_1=0.142857$$

$$W_2=0.357143$$

$$W_3= 0.5$$

➤ **Separate Type Ratio Estimator**

$$\begin{aligned}\bar{y}_{rsst} &= \sum_{h=1}^3 W_h \frac{\bar{y}_h}{\bar{x}_h} \bar{X}_h \\ \bar{y}_{rsst} &= \sum_{h=1}^3 W_h \bar{y}_{rh} \\ &= W_1 \bar{y}_{r1} + W_2 \bar{y}_{r2} + W_3 \bar{y}_{r3} \\ &= 0.143 \times 2.55 + 0.357 \times 9.12 \\ &\quad + 0.5 \times 20.65 \\ &= 13.95\end{aligned}$$

Q. Find the variance expression of combined Type Regression Estimator?

Answer:

➤ **Regression Estimator**

- The Regression Estimator without stratification

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx}(\bar{X} - \bar{x})$$

Regression Estimator $\hat{\beta}_{yx} = \frac{S_{yx}}{S_x^2}$ mcq's

➤ **Combined Type Regression Estimator**

- The Regression Estimator under stratified sampling

$$\bar{y}_{regc} = \bar{y}_{st} + \hat{\beta}_{yx}(\bar{X} - \bar{x}_{st})$$

$$\bar{y}_{regc} = \sum_{h=1}^k W_h \bar{y}_h + \hat{\beta}_{yx}(\bar{X} - \sum_{h=1}^k W_h \bar{x}_h)$$

➤ **Expected Value of Estimator**

- Taking Expected Value on both sides

$$E(\bar{y}_{regc}) = \sum_{h=1}^k W E(\bar{y}_h) + \hat{\beta}_{yx} \left(\bar{X} - \sum_{h=1}^k W_h E(\bar{x}_h) \right)$$

$$E(\bar{y}_{regc}) = \sum_{h=1}^k W \bar{Y} + \hat{\beta}_{yx} \left(\bar{X} - \sum_{h=1}^k W_h \bar{X} \right)$$

$$E(\bar{y}_{regc}) = \sum_{h=1}^k W \bar{Y} + \hat{\beta}_{yx} \left(\bar{X} - \sum_{h=1}^k W_h \bar{X} \right)$$

➤ **Expectation of Regression Estimator**

$$E(\bar{y}_{regc}) = \bar{Y} + \hat{\beta}_{yx}(\bar{X} - \bar{X})$$

$$E(\bar{y}_{regc}) = \bar{Y}$$

imp short

➤ **Variance of Regression Estimator**

$$\bar{y}_{regc} = \bar{y}_{st} + \hat{\beta}_{yx}(\bar{X} - \bar{x}_{st})$$

$$V(\bar{y}_{regc}) = E[\bar{y}_{regc} - \bar{Y}]^2$$

$$= E \left[\bar{y}_{st} + \hat{\beta}_{yx}(\bar{X} - \bar{x}_{st}) - \bar{Y} \right]^2$$

$$V(\bar{y}_{regc}) = E \left[\bar{y}_{st} + \hat{\beta}_{yx}(\bar{X} - \bar{x}_{st}) - \bar{Y} \right]^2$$

$$V(\bar{y}_{regc}) = E \left[\sum_{h=1}^k W_h \bar{y}_h + \hat{\beta}_{yx}(\bar{X} - \sum_{h=1}^k W_h \bar{x}_h) - \bar{Y} \right]^2$$

$$V(\bar{y}_{regc}) = E \left[\sum_{h=1}^k W_h (\bar{y}_h - \bar{Y}) - \hat{\beta}_{yx} \sum_{h=1}^k W_h (\bar{x}_h - \bar{X}) \right]^2$$

$$V(\bar{y}_{regc}) = E \left[\sum_{h=1}^k W_h (\bar{y}_h - \bar{Y}) - \hat{\beta}_{yx} \sum_{h=1}^k W_h (\bar{x}_h - \bar{X}) \right]^2$$

➤ **Variance of Combined Type Regression Estimator**

$$V(\bar{y}_{regc}) = \sum_{h=1}^k W_h^2 \lambda_h \begin{bmatrix} S_{hy}^2 + \beta_0^2 S_{hx}^2 \\ -2\beta_0 S_{hxy} \end{bmatrix}$$

$$\beta_0 = \frac{Cov(\bar{x}_h, \bar{y}_h)}{Var(\bar{x}_h)}$$

$$= \frac{\sum_{i=1}^k \frac{w_i^2 f_i}{n_i} S_{iXY}}{\sum_{i=1}^k \frac{w_i^2 f_i}{n_i} S_{iX}^2}$$

ie minimum variance is given by

$$Var_{min}(\hat{\bar{y}}_{regc}) = \sum_{i=1}^k \frac{w_i^2 f_i}{n_i} (S_{iY}^2 - \beta_0^2 S_{iX}^2).$$

Q. Find the variance of Separate Type Regression Estimator?

Answer:

➤ **Regression Estimator**

- The Regression Estimator without stratification

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx} (\bar{X} - \bar{x})$$

$$\hat{\beta}_{yx} = \frac{S_{yx}}{S_x^2}$$

➤ **Separate Type Regression Estimator**

- The Combined Type Regression Estimator is

$$\bar{y}_{regc} = \bar{y}_{st} + \hat{\beta}_{yx} (\bar{X} - \bar{x}_{st}) \quad \text{imp mcq's}$$

- The Separate Type Regression Estimator is

$$\bar{y}_{regs} = \sum_{h=1}^k W_h \left[\bar{y}_h + \hat{\beta}_{yxh} (\bar{X} - \bar{x}_h) \right]$$

➤ **Expectation of Separate Type Regression Estimator**

$$\begin{aligned}
 E(\bar{y}_{regs}) &= \sum_{h=1}^k W_h \left[E(\bar{y}_h) + \hat{\beta}_{yxh} (\bar{X}_h - E(\bar{x}_h)) \right] \\
 E(\bar{y}_{regs}) &= \sum_{h=1}^k W_h \left[\bar{Y}_h + \hat{\beta}_{yxh} (\bar{X}_h - \bar{X}_h) \right] \\
 E(\bar{y}_{regs}) &= \sum_{h=1}^k W_h \bar{Y}_h = \bar{Y}
 \end{aligned}$$

➤ **Variance of Regression Estimator**

$$\begin{aligned}
 \bar{y}_{regs} &= \sum_{h=1}^k W_h \bar{y}_h + \hat{\beta}_{yxh} \sum_{h=1}^k W_h (\bar{X}_h - \bar{x}_h) \\
 V(\bar{y}_{regs}) &= E[\bar{y}_{regs} - \bar{Y}]^2 \\
 V(\bar{y}_{regs}) &= E \left[\sum_{h=1}^k W_h \bar{y}_h + \hat{\beta}_{yxh} \sum_{h=1}^k W_h (\bar{X}_h - \bar{x}_h) - \bar{Y} \right]^2 \\
 V(\bar{y}_{regs}) &= E \left[\sum_{h=1}^k W_h (\bar{y}_h - \bar{Y}) - \sum_{h=1}^k W_h \hat{\beta}_{yxh} (\bar{x}_h - \bar{X}_h) \right]^2 \\
 V(\bar{y}_{regs}) &= \left[\sum_{h=1}^k W_h^2 E(\bar{y}_h - \bar{Y})^2 + \sum_{h=1}^k W_h^2 \hat{\beta}_{yxh}^2 E(\bar{x}_h - \bar{X}_h)^2 \right. \\
 &\quad \left. - 2 \sum_{h=1}^k W_h^2 \hat{\beta}_{yxh} E(\bar{y}_h - \bar{Y})(\bar{x}_h - \bar{X}_h) \right] \\
 V(\bar{y}_{regs}) &= \sum_{h=1}^k W_h^2 \lambda_h \begin{bmatrix} S_{hy}^2 + \beta_{yxh}^2 S_{hx}^2 \\ -2\beta_{yxh} S_{hxy} \end{bmatrix} \\
 \beta_{0i} &= \frac{S_{ixy}}{S_{ix}^2}
 \end{aligned}$$

Q. Simulation Study for Combined Type Ratio Estimator.

Answer:

➤ **Simulation Study**

```
library(mvtnorm)
#population and sample size.
N=300;N1=100;N2=100;N3=100;n1=15;n2=15;n3=15;n=45;
```



```
w1=N1/N;
w2=N2/N;
w3=N3/N;
```

imp mcq's

➤ Defining Parameters

```
# mean vectors for stratum 1, 2 and 3.
m1=c(10,5);
m2=c(20,10);
m3=c(30,15)
#variance covariance matrix for #stratum 1, 2 and 3 given #below.
sig1=matrix(c(1,0.85,0.85,1),ncol=2);
sig2=matrix(c(1,0.80,0.80,1),ncol=2); sig3=matrix(c(1,0.75,0.75,1),ncol=2);
```

s/q

➤ Generating Populations

```
r1=rmvnorm(N1,m1,sig1);
r2=rmvnorm(N2,m2,sig2);
r3=rmvnorm(N3,m3,sig3);
# generate random variable for #stratum 1.
y1=r1[,1];x1=r1[,2]
# generate random variable for #stratum 2.
y2=r2[,1];x2=r2[,2]
# generate random variable for #stratum 3.
y3=r3[,1];x3=r3[,2]
x<-c(x1,x2,x3)
```

➤ Looping

```
for(i in 1:5000){
  sa1=sample(1:N1,n1)
  xs1=x1[sa1];ys1=y1[sa1];
  sa2=sample(1:N2,n2)
  xs2=x2[sa2];ys2=y2[sa2];
  sa3=sample(1:N3,n3)
  xs3=x3[sa3];ys3=y3[sa3];
  yst[i]=w1*mean(ys1)+w2*mean(ys2)+w3*mean(ys3);
  xst[i]=w1*mean(xs1)+w2*mean(xs2)+w3*mean(xs3);
  rst[i]=yst[i]*(mean(x)/(xst[i]))
}
mean(yst);mean(xst);
mean(rst);
var(yst);var(rst);
var(yst)/var(rst)
```

➤ Means and Variances

```
> mean(yst);mean(xst);mean(rst);
[1] 20.01764
[1] 10.02838
[1] 20.02706
```



```
> var(yst);var(rst);var(yst)/var(rst)
```

Q. Define the process of simulation study for combined type ratio estimator also find mean and variance of estimator using following data

```
N=300;N1=100;N2=100;N3=100;n1=15;n2=15;n3=15;n=45;
```

```
# mean vectors for stratum 1, 2 and 3.
```

```
m1=c(10,5);
```

```
m2=c(20,10);
```

```
m3=c(30,15)
```

```
#variance covariance matrix for #stratum 1, 2 and 3 given below.
```

```
sig1=matrix(c(1,0.85,0.85,1),ncol=2)
```

```
sig2=matrix(c(1,0.80,0.80,1),ncol=2); sig3=matrix(c(1,0.75,0.75,1),ncol=2);
```

Ans.

```
library(mvtnorm)
```

```
#population and sample size.
```

```
N=300;N1=100;N2=100;N3=100;n1=15;n2=15;n3=15;n=45;
```

```
w1=N1/N;
```

```
w2=N2/N;
```

```
w3=N3/N;
```

```
# mean vectors for stratum 1, 2 and 3.
```

```
m1=c(10,5);
```

```
m2=c(20,10);
```

```
m3=c(30,15)
```

```
#variance covariance matrix for #stratum 1, 2 and 3 given below.
```

```
sig1=matrix(c(1,0.85,0.85,1),ncol=2)
```

```
sig2=matrix(c(1,0.80,0.80,1),ncol=2); sig3=matrix(c(1,0.75,0.75,1),ncol=2);
```

```
r1=rmvnorm(N1,m1,sig1);
```

```
r2=rmvnorm(N2,m2,sig2);
```

```
r3=rmvnorm(N3,m3,sig3);
```

generate random variable for #stratum 1.

```
y1=r1[,1];x1=r1[,2]
```

generate random variable for #stratum 2

```
y2=r2[,1];x2=r2[,2]
```

generate random variable for #stratum 3.

```
y3=r3[,1];x3=r3[,2]
```

```
x<-c(x1,x2,x3)
```

```
for(i in 1:5000){
```

```
sa1=sample(1:N1,n1)
```

```
xs1=x1[sa1];ys1=y1[sa1];
```

```
sa2=sample(1:N2,n2)
```

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feb 2024


```
xs2=x2[sa2];ys2=y2[sa2];
sa3=sample(1:N3,n3)
xs3=x3[sa3];ys3=y3[sa3];
yst[i]=w1*mean(ys1)+w2*mean(ys2)+w3*mean(ys3);
xst[i]=w1*mean(xs1)+w2*mean(xs2)+w3*mean(xs3);
rst[i]=yst[i]*(mean(x)/(xst[i]))
}
mean(yst);mean(xst);
mean(rst);
var(yst);var(rst);
var(yst)/var(rst)
```

#Means and Variances

```
> mean(yst);mean(xst);mean(rst);
> var(yst);var(rst);var(yst)/var(rst)
```

Q. Define the process of simulation study for separate type ratio estimator also find mean and variance of estimator using following data

```
library(mvtnorm)
#population and sample size.
N=300;N1=100;N2=100;N3=100;n1=15;n2=15;n3=15;n=45;
# mean vectors for stratum 1, 2 and 3.
m1=c(10,5);
m2=c(20,10);
m3=c(30,15)
#variance covariance matrix for #stratum 1, 2 and 3 given #below.
sig1=matrix(c(1,0.85,0.85,1),ncol=2);
sig2=matrix(c(1,0.80,0.80,1),ncol=2); sig3=matrix(c(1,0.75,0.75,1),ncol=2);
```

Ans:

```
library(mvtnorm)
#population and sample size.
N=300;N1=100;N2=100;N3=100;n1=15;n2=15;n3=15;n=45;
w1=N1/N;
w2=N2/N;
w3=N3/N;

# mean vectors for stratum 1, 2 and 3.
m1=c(10,5);
m2=c(20,10);
m3=c(30,15)

#variance covariance matrix for #stratum 1, 2 and 3 given #below.
sig1=matrix(c(1,0.85,0.85,1),ncol=2);
sig2=matrix(c(1,0.80,0.80,1),ncol=2); sig3=matrix(c(1,0.75,0.75,1),ncol=2);
r1=rmvnorm(N1,m1,sig1);
```



```
r2=rmvnorm(N2,m2,sig2);
r3=rmvnorm(N3,m3,sig3);

# generate random variable for #stratum 1.
y1=r1[,1];x1=r1[,2]

# generate random variable for #stratum 2.
y2=r2[,1];x2=r2[,2]

# generate random variable for #stratum 3.
y3=r3[,1];x3=r3[,2]
x<-c(x1,x2,x3)
for(i in 1:5000){
sa1=sample(1:N1,n1)
xs1=x1[sa1];ys1=y1[sa1];
r1[i]=mean(ys1)*(mean(x1)/mean(xs1))
sa2=sample(1:N2,n2)
xs2=x2[sa2];ys2=y2[sa2];
r2[i]=mean(ys2)*(mean(x2)/mean(xs2))
sa3=sample(1:N3,n3)
xs3=x3[sa3];ys3=y3[sa3];
r3[i]=mean(ys3)*(mean(x3)/mean(xs3))
rsst[i]=w1*r1[i]+w2*r2[i]+w3*r3[i];
yst[i]=w1*mean(ys1)+w2*mean(ys2)+w3*mean(ys3);
}
mean(rsst);mean(yst);
var(yst);var(rsst);var(yst)/var(rsst)
```

Means and Variances

```
> mean(rsst);mean(yst);
```


Double Sampling

Most imp lecture

Introduction:

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Ratio and regression methods of estimation require the knowledge of population mean of auxiliary variable in advance. An estimate of mean of auxiliary variable from a large sample may be used. This procedure of selecting a large sample for collecting information on auxiliary variable x and then selecting a subsample from it for collecting the information on the study variable y is called double sampling or two-phase sampling.

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The estimate is calculated by taking the sample in two phases. The expected value of the statistic is

$$E(t) = E_1(E_2(t))$$

The variance of the statistic is

$$Var(t) = E(t - E(t))^2$$

$$Var(t) = E[(t - E_2(t)) + (E_2(t) - E(t))]^2$$

$$Var(t) = E(t - E_2(t))^2 + E(E_2(t) - E(t))^2$$

$$Var(t) = E_1 E_2(t - E_2(t))^2 + E_1 E_2(E_2(t) - E(t))^2$$

$$Var(t) = E_1[V_2(t)] + E_1(E_2(t) - E_1(E_2(t)))^2$$

$$Var(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

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The Ratio Estimator for population mean

$$\bar{y}_r = \frac{\bar{y}}{\bar{x}} \bar{X}$$

The Ratio Estimator for Double Sampling

$$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$$

Notations

$$e_0 = (\bar{y}_2 - \bar{Y}) / \bar{Y}$$

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2024

$$e_1 = (\bar{x}_1 - \bar{X}) / \bar{X}$$

$$e_2 = (\bar{x}_2 - \bar{X}) / \bar{X}$$

Using these notations

$$E(e_0) = 0, E(e_0^2) = \lambda_2 C_y^2,$$

$$E(e_1) = 0, E(e_1^2) = \lambda_1 C_x^2,$$

$$E(e_2) = 0, E(e_2^2) = \lambda_2 C_x^2,$$

$$E(e_1 e_2) = \lambda_1 C_x^2,$$

$$E(e_0 e_1) = \lambda_1 \rho_{yx} C_y C_x, E(e_0 e_2) = \lambda_2 \rho_{yx} C_y C_x,$$

Prove that $E(e_1 e_2) = \lambda_1 C_x^2$

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must paper

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Proof:

$$e_1 e_2 = \frac{(\bar{x}_1 - \bar{X})(\bar{x}_2 - \bar{X})}{\bar{X}^2} \Rightarrow E(e_1 e_2) = \frac{E(\bar{x}_1 - \bar{X})(\bar{x}_2 - \bar{X})}{\bar{X}^2}$$

$$E(e_1 e_2) = \frac{E_1 E_2 (\bar{x}_1 - \bar{X})(\bar{x}_2 - \bar{X})}{\bar{X}^2}$$

$$E(e_1 e_2) = \frac{E_1 (\bar{x}_1 - \bar{X})(\bar{x}_1 - \bar{X})}{\bar{X}^2}$$

$$E(e_1 e_2) = \frac{E_1 (\bar{x}_1 - \bar{X})^2}{\bar{X}^2}$$

$$E(e_1 e_2) = \frac{\lambda_1 S_x^2}{\bar{X}^2}$$

$$E(e_1 e_2) = \lambda_1 C_x^2$$

Q. Find Bias of Ratio Estimator in Double Sampling.

The Ratio Estimator for Double Sampling

$$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$$

$$e_0 = (\bar{y}_2 - \bar{Y}) / \bar{Y}$$

$$e_1 = (\bar{x}_1 - \bar{X}) / \bar{X}$$

$$e_2 = (\bar{x}_2 - \bar{X}) / \bar{X}$$

$$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$$

$$\bar{y}_{rd} = \frac{\bar{Y}(1+e_0)}{\bar{X}(1+e_2)} \bar{X}(1+e_1)$$

$$\bar{y}_{rd} = \frac{\bar{Y}(1+e_0)}{(1+e_2)} (1+e_1)$$

The Bias of Ratio Estimator in Double Sampling

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1)(1+e_2)^{-1}$$

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1)(1-e_2+e_2^2-....)$$

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1)(1-e_2+e_2^2)$$

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1-e_2-e_1e_2+e_2^2+e_1e_2^2)$$

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1-e_2-e_1e_2+e_2^2)$$

$$\bar{y}_{rd} - \bar{Y} = \bar{Y}(e_1 - e_2 - e_1e_2 + e_2^2 + e_0 + e_0e_1 - e_0e_2)$$

$$E(\bar{y}_{rd} - \bar{Y}) = \bar{Y}E(e_1 - e_2 - e_1e_2 + e_2^2 + e_0 + e_0e_1 - e_0e_2)$$

$$E(\bar{y}_{rd} - \bar{Y}) = \bar{Y}(\lambda_2 C_x^2 - \lambda_1 C_x^2 + \lambda_1 \rho C_y C_x - \lambda_2 \rho C_y C_x)$$

$$E(\bar{y}_{rd} - \bar{Y}) = \bar{Y} [(\lambda_2 - \lambda_1)C_x^2 - (\lambda_2 - \lambda_1)\rho C_y C_x]$$

$$E(\bar{y}_{rd} - \bar{Y}) = \bar{Y} (\lambda_2 - \lambda_1) [C_x^2 - \rho C_y C_x]$$

$$\text{Bias}(\bar{y}_{rd}) = \bar{Y} (\lambda_2 - \lambda_1) [C_x^2 - \rho C_y C_x]$$

Q. Find MSE of Ratio Estimator in Double Sampling.

The Ratio Estimator for Double Sampling

$$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$$

$$e_0 = (\bar{y}_2 - \bar{Y}) / \bar{Y}$$

$$e_1 = (\bar{x}_1 - \bar{X}) / \bar{X}$$

$$e_2 = (\bar{x}_2 - \bar{X}) / \bar{X}$$

most imp L/q

$$\bar{y}_{rd} = \frac{\bar{y}_2}{\bar{x}_2} \bar{x}_1$$

past paper

$$\bar{y}_{rd} = \frac{\bar{Y}(1+e_0)}{\bar{X}(1+e_2)} \bar{X}(1+e_1)$$

$$\bar{y}_{rd} = \frac{\bar{Y}(1+e_0)}{(1+e_2)} (1+e_1)$$

MSE of Ratio Estimator in Double Sampling

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1)(1+e_2)^{-1}$$

$$\bar{y}_{rd} = \bar{Y}(1+e_0)(1+e_1)(1-e_2+e_2^2-....)$$

$$\bar{y}_{rd} - \bar{Y} = \bar{Y}(e_1 - e_2 + e_0)$$

$$E(\bar{y}_{rd} - \bar{Y})^2 = \bar{Y}^2 E(e_1 - e_2 + e_0)^2$$

$$E(\bar{y}_{rd} - \bar{Y})^2 = \bar{Y}^2 E(e_1^2 + e_2^2 + e_0^2 + e_0e_1 - e_0e_2 - e_1e_2)$$

$$MSE(\bar{y}_{rd}) = \bar{Y}^2 [\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 - 2C_x C_y \rho_{xy})]$$

$$MSE(\bar{y}_{rd}) = \lambda_2 \bar{Y}^2 (C_y^2 + C_x^2 - 2C_x C_y \rho_{xy}) - \lambda_1 \bar{Y}^2 (C_x^2 - 2C_x C_y \rho_{xy})$$

Q. Find Bias of Product Estimator in Double Sampling.

The Product Estimator is

$$\bar{y}_p = \bar{y} \frac{\bar{x}}{\bar{X}}$$

The Product Estimator in Double Sampling is

$$\bar{y}_{pd} = \frac{\bar{y}_2 \bar{x}_2}{\bar{x}_1}$$

Double sampling all questions very important in short and long question and repeated in past and current papers

$$e_0 = (\bar{y}_2 - \bar{Y})/\bar{Y}, e_1 = (\bar{x}_1 - \bar{X})/\bar{X}, e_2 = (\bar{x}_2 - \bar{X})/\bar{X}$$

$$\bar{y}_2 = \bar{Y}(1+e_0), \bar{x}_1 = \bar{X}(1+e_1), \bar{x}_2 = \bar{X}(1+e_2)$$

Notations

$$E(e_0) = 0, E(e_0^2) = \lambda_2 C_y^2, E(e_1) = 0$$

$$E(e_1^2) = \lambda_1 C_x^2, E(e_1) = 0, E(e_2^2) = \lambda_2 C_x^2$$

$$E(e_1 e_2) = \lambda_1 C_x^2, E(e_0 e_1) = \lambda_1 \rho_{yx} C_y C_x, E(e_0 e_2) = \lambda_2 \rho_{yx} C_y C_x$$

Bias of Product Estimator in Double Sampling

$$\bar{y}_{pd} = \frac{\bar{y}_2}{\bar{x}_1} \bar{x}_2$$

$$\bar{y}_{pd} = \frac{\bar{Y}(1+e_0)}{\bar{X}(1+e_1)} \bar{X}(1+e_2)$$

$$\bar{y}_{pd} = \frac{\bar{Y}(1+e_0)}{(1+e_1)} (1+e_2)$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2)(1+e_1)^{-1}$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2)(1-e_1+e_1^2-....)$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2)(1-e_1+e_1^2)$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2-e_1-e_1e_2+e_1^2+e_2e_1^2)$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2-e_1-e_1e_2+e_1^2)$$

$$\bar{y}_{pd} - \bar{Y} = \bar{Y}(e_2 - e_1 - e_1e_2 + e_1^2 + e_0 + e_0e_2 - e_0e_1)$$

$$E(\bar{y}_{pd} - \bar{Y}) = \bar{Y}E(e_2 - e_1 - e_1e_2 + e_1^2 + e_0 + e_0e_2 - e_0e_1)$$

$$E(\bar{y}_{pd} - \bar{Y}) = \bar{Y}(\lambda_1 C_x^2 - \lambda_1 C_x^2 + \lambda_2 \rho C_y C_x - \lambda_1 \rho C_y C_x)$$

$$Bias(\bar{y}_{pd}) = \bar{Y}[(\lambda_2 - \lambda_1)\rho C_y C_x]$$

Q. Find MSE of Product Estimator in Double Sampling.

The Product Estimator for Double Sampling

$$\bar{y}_{pd} = \frac{\bar{y}_2}{\bar{x}_1} \bar{x}_2$$

$$e_0 = (\bar{y}_2 - \bar{Y})/\bar{Y}, e_1 = (\bar{x}_1 - \bar{X})/\bar{X}, e_2 = (\bar{x}_2 - \bar{X})/\bar{X}$$

$$\bar{y}_{pd} = \frac{\bar{y}_2}{\bar{x}_1} \bar{x}_2, \bar{y}_{pd} = \frac{\bar{Y}(1+e_0)}{\bar{X}(1+e_1)} \bar{X}(1+e_2), \bar{y}_{pd} = \frac{\bar{Y}(1+e_0)}{(1+e_1)}(1+e_2)$$

The MSE of Product Estimator in Double Sampling

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2)(1+e_1)^{-1}$$

$$\bar{y}_{pd} = \bar{Y}(1+e_0)(1+e_2)(1-e_1+....)$$

$$\bar{y}_{pd} - \bar{Y} = \bar{Y}(e_2 - e_1 + e_0)$$

imp s/q

$$E(\bar{y}_{pd} - \bar{Y})^2 = \bar{Y}^2 E(e_2 - e_1 + e_0)^2$$

$$E(\bar{y}_{pd} - \bar{Y})^2 = \bar{Y}^2 E(e_1^2 + e_2^2 + e_0^2 + 2e_0e_2 - 2e_0e_1 - 2e_1e_2)$$

$$MSE(\bar{y}_{rd}) = \bar{Y}^2 \left[\lambda_2 C_y^2 + (\lambda_2 - \lambda_1)(C_x^2 + 2C_x C_y \rho_{xy}) \right]$$

$$MSE(\bar{y}_{rd}) = \lambda_2 \bar{Y}^2 (C_y^2 + C_x^2 + 2C_x C_y \rho_{xy}) - \lambda_1 \bar{Y}^2 (C_x^2 + 2C_x C_y \rho_{xy})$$

Q. Generate a population of size 1000 for the given parameters using bivariate normal distribution, such that

$$[Y, X] \text{ as } \mu = [10 \ 2].$$

$$\Sigma = \begin{bmatrix} 1 & 0.85 \\ 0.85 & 1 \end{bmatrix}$$

s/q

Consider sample sizes n1=100;n2=20. Select 10,000 random samples considering double and calculate

$$t = \bar{y}_2 \frac{\bar{x}_1}{\bar{x}_2}$$

Further, calculate the MSEs for above estimator.

Ans:

```
library(mvtnorm)
```

```
N=1000;ryx=0.85; n1=100;n2=20;
```

```
m=c(10,2); # vector of mean.
```

```
# variance covariance matrix is given below.
```

```
sig=matrix(c(1,0.85,0.85,1),ncol=2);
```

```
r=rmvnorm(N,m,sig);
```

```
x=r[,2];y=r[,1];
```

```
r<-c();mratio<-c();m=c();
```

```
for(i in 1:10000)
```

```
{
```

```
s1<-sample(1:N,n1)
```

```
s2<-sample(s1,n2)
```

```
m[i]<-mean(x[s1])
```

```
r[i] <- mean(y[s2])/mean(x[s2])
```

```
mratio[i] <- r[i]*m[i]
```

```
}
```

```
mean(mratio)
```

```
var(mratio)
```

repeated all this type of questions in mcq's and short question

Two Stage Sampling

Q. Prove that sample mean is unbiased estimator of population mean in two stage sampling

$$\bar{\bar{y}} = \sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm}$$

Ans:

The estimate is calculated by taking the sample in two stages

The expected value and variance of the statistic is

$$E(t) = E_1(E_2(t))$$

$$Var(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

Expected Value of Mean

$$\bar{\bar{y}} = \sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm}$$

$$E(\bar{\bar{y}}) = E_1(E_2(\bar{\bar{y}}))$$

$$E(\bar{\bar{y}}) = E_1\left(E_2\left(\sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(\sum_{i=1}^n \frac{1}{n} E_2\left(\sum_{j=1}^m \frac{y_{ij}}{m}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(\sum_{i=1}^n \frac{1}{n} \left(\sum_{j=1}^m \frac{E_2(y_{ij})}{m}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(\sum_{i=1}^n \frac{1}{n} \left(\sum_{j=1}^M \frac{y_{ij}}{M}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(\sum_{i=1}^n \frac{1}{n} \bar{Y}_i\right)$$

$$E(\bar{\bar{y}}) = \left(\sum_{i=1}^n \frac{1}{n} E_1(\bar{Y}_i)\right)$$

$$E(\bar{\bar{y}}) = E_1(\bar{Y}_i)$$

$$E(\bar{y}) = \left(\sum_{i=1}^N \frac{\bar{Y}_i}{N} \right) \Rightarrow E(\bar{y}) = \bar{Y}$$

Q. Prove that sample mean is unbiased estimator of population mean in two stage sampling.

$$\bar{y} = \sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm}$$

Ans:

$$\bar{y} = \sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm} = \sum_{i=1}^n \frac{1}{n} \bar{y}_i$$

$$Var(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

$$Var(\bar{y}) = E_1[V_2(\bar{y})] + V_1[E_2(\bar{y})]$$

$$E_2(\bar{y}) = E_2 \left[\sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm} \right]$$

$$E_2(\bar{y}) = \left[\sum_{i=1}^n \frac{1}{n} \sum_{j=1}^m \frac{E_2(y_{ij})}{m} \right]$$

$$E_2(\bar{y}) = \left[\sum_{i=1}^n \frac{1}{n} \sum_{j=1}^M \frac{y_{ij}}{M} \right]$$

$$E_2(\bar{y}) = \left[\sum_{i=1}^n \frac{1}{n} \sum_{j=1}^M \frac{y_{ij}}{M} \right]$$

Read it same method hai

$$E_2(\bar{y}) = \left[\sum_{i=1}^n \frac{1}{n} \bar{Y}_i \right]$$

$$E_2(\bar{y}) = \bar{Y}_n$$

$\bar{Y}_n$ is the estimator based on the 1st stage sample of size n, by SRS, we have

$$V(\bar{y}) = \left(\frac{N-n}{Nn} \right) S^2$$

$$V_1[E_2(\bar{y})] = V_1[\bar{Y}_n] \Rightarrow V_1[E_2(\bar{y})] = \left(\frac{N-n}{Nn} \right) S_1^2$$

$$E_1[V_2(\bar{y})] = E_1 \left[V_2 \left(\sum_{i=1}^n \frac{1}{n} \bar{y}_i \right) \right]$$

$$E_1[V_2(\bar{y})] = E_1 \left[\frac{1}{n^2} \sum_{i=1}^n V_2(\bar{y}_i) + \frac{1}{n^2} \sum_{i \neq j=1}^n \sum Cov(\bar{y}_i, \bar{y}_j) \right]$$

$$E_1[V_2(\bar{y})] = E_1 \left[\frac{1}{n^2} \sum_{i=1}^n V_2(\bar{y}_i) \right] = \frac{1}{n^2} E_1 \left[\sum_{i=1}^n \frac{M-m}{Mm} S_{2i}^2 \right]$$

$$E_1[V_2(\bar{y})] = \frac{1}{n^2} \left[\sum_{i=1}^n \frac{M-m}{Mm} E_1(S_{2i}^2) \right] = \frac{1}{n} \left[\frac{M-m}{Mm} E_1(S_{2i}^2) \right]$$

$$E_1[V_2(\bar{y})] = \frac{1}{n} \left[\frac{M-m}{Mm} \sum_{i=1}^N \frac{S_{2i}^2}{N} \right]$$

$$E_1[V_2(\bar{y})] = \frac{1}{n} \left[\frac{M-m}{Mm} S_2^2 \right]$$

$$Var(\bar{y}) = E_1[V_2(\bar{y})] + V_1[E_2(\bar{y})]$$

$$E_1[V_2(\bar{y})] = \frac{1}{n} \left[\frac{M-m}{Mm} S_2^2 \right]$$

$$V_1[E_2(\bar{y})] = \left(\frac{N-n}{Nn} \right) S_1^2$$

$$Var(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

$$Var(t) = \frac{1}{n} \left[\frac{M-m}{Mm} S_2^2 \right] + \left(\frac{N-n}{Nn} \right) S_1^2$$

$$Var(t) = \frac{1}{n} \left[\frac{M-m}{Mm} s_2^2 \right] + \left(\frac{N-n}{Nn} \right) s_1^2$$

Q. Explain two stage sampling for unequal first stage units, also find expected value of following estimator.

$$\bar{\bar{y}} = \sum_{i=1}^n \frac{M_i \bar{y}_i}{n\bar{M}} = \sum_{i=1}^n \frac{u_i \bar{y}_i}{n}$$

Let the population has N clusters with M_i as size of i^{th} cluster. Let a sample of n first stage units be selected from this population. Let m_i units will be selected at second stage.
 M_i = Size of i^{th} cluster

$$M_0 = \sum_{i=1}^N M_i = N\bar{M}$$

imp mcq's

Mean of i^{th} first stage unit

$$\bar{Y}_i = \frac{\sum_{j=1}^{M_i} y_{ij}}{M_i}$$

Overall population mean

$$\bar{Y} = \frac{\sum_{i=1}^N M_i \bar{Y}_i}{N\bar{M}}$$

The Expected Value of Mean

$$\bar{\bar{y}} = \sum_{i=1}^n \frac{M_i \bar{y}_i}{n\bar{M}} = \sum_{i=1}^n \frac{u_i \bar{y}_i}{n}$$

$$E(\bar{\bar{y}}) = E_1(E_2(\bar{\bar{y}}))$$

$$E(\bar{\bar{y}}) = E_1\left(E_2\left(\sum_{i=1}^n \frac{u_i \bar{y}_i}{n}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(E_2\left(\sum_{i=1}^n \frac{u_i \bar{y}_i}{n}\right)\right)$$

$$E(\bar{\bar{y}}) = E_1\left(\sum_{i=1}^n \frac{u_i \bar{Y}_i}{n}\right)$$

$$E(\bar{\bar{y}}) = \left(\sum_{i=1}^n \frac{M_i E_1(\bar{Y}_i)}{n} \right)$$

$$E(\bar{\bar{y}}) = \left(\sum_{i=1}^n \frac{E_1(u_i \bar{Y}_i)}{n} \right)$$

$$E(\bar{\bar{y}}) = \left(\sum_{i=1}^N \frac{u_i \bar{Y}_i}{N} \right)$$

$$E(\bar{\bar{y}}) = \left(\sum_{i=1}^N \frac{M_i \bar{Y}_i}{\bar{M}N} \right)$$

$$E(\bar{\bar{y}}) = \bar{Y}$$

Q. Find the variance expression of following estimator in two stage sampling when first stage units are not equal in sizes.

$$\bar{\bar{y}} = \sum_{i=1}^n \frac{M_i \bar{y}_i}{n\bar{M}} = \sum_{i=1}^n \frac{u_i \bar{y}_i}{n}$$

Ans

$$Var(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

$$Var(\bar{\bar{y}}) = E_1[V_2(\bar{\bar{y}})] + V_1[E_2(\bar{\bar{y}})]$$

$$V_1[E_2(\bar{\bar{y}})] = V_1\left[E_2\left(\sum_{i=1}^n \frac{u_i \bar{y}_i}{n}\right)\right]$$

$$V_1[E_2(\bar{\bar{y}})] = V_1\left[\sum_{i=1}^n \frac{u_i \bar{Y}_i}{n}\right]$$

$$V_1[E_2(\bar{\bar{y}})] = V_1[\bar{y}]$$

$$V_1[E_2(\bar{\bar{y}})] = \left(\frac{1}{n} - \frac{1}{N}\right) S_1^2$$

$$E_1[V_2(\bar{\bar{y}})] = E_1\left[V_2\left(\sum_{i=1}^n \frac{u_i \bar{y}_i}{n}\right)\right]$$

$$E_1[V_2(\bar{y})] = E_1\left[\sum_{i=1}^n \frac{u_i^2}{n^2} V_2(\bar{y}_i)\right] = E_1\left[\sum_{i=1}^n \frac{u_i^2}{n^2} \frac{M_i - m_i}{M_i m_i} S_{2i}^2\right]$$

$$E_1[V_2(\bar{y})] = \sum_{i=1}^n \frac{u_i^2}{n^2} \frac{M_i - m_i}{M_i m_i} = \sum_{i=1}^N \frac{u_i^2}{n} \frac{M_i - m_i}{M_i m_i} \frac{S_{2i}^2}{N}$$

$$\text{Var}(t) = E_1[V_2(t)] + V_1[E_2(t)]$$

$$\text{Var}(t) = \sum_{i=1}^N \frac{u_i^2}{n} \frac{M_i - m_i}{M_i m_i} \frac{S_{2i}^2}{N} + \left(\frac{N-n}{Nn}\right) S_1^2$$

Short question 1 ya 2 cluters dy gy long main sab R programing likhni hai.....!

Q. Write the program for Two Stage Sampling using R language considering following data taking n1=3;n2=2. Calculate the sample mean

Cluster-1	125	115	129	134	111
Cluster-2	134	125	142	141	131
Cluster-3	144	143	122	134	126
Cluster-4	114	111	134	131	146
Cluster-5	119	126	122	129	130
Cluster-6	140	125	124	124	115

Ans:

#--Defining Clusters in R----

Clu1<-c(125,115,129,134,111)

slq

Clu2<-c(134,125,142,141,131)

Clu3<-c(144,143,122,134,126)

Clu4<-c(114,111,134,131,146)

Clu5<-c(119,126,122,129,130)

Clu6<-c(140,125,124,124,115)

Clus<-data.frame(Clus1,Clus2,Clus3,Clus4,Clus5,Clus6)

pop<-t(Clus)

1st Stage Sample

n1=3;n2=2;

w<-pop[sample(nrow(pop),n1),]

sc1<-w[1,]

sc2<-w[2,]


```
sc3<-w[3,]
s2_sc1<-sample(sc1,n2)
s2_sc2<-sample(sc2,n2)
s2_sc3<-sample(sc3,n2)
f_s=c(s2_sc1,s2_sc2,s2_sc3)
est=mean(f_s)
```

Q2: Generate the three clusters such that

Cluster-1 from normal distribution with mean zero and variance 1.

Cluster-2 from normal distribution with mean 2 and variance 5.

Cluster-3 from normal distribution with mean 4 and variance 9.

Calculate the sample mean using two stage cluster sampling taking $n_1=2; n_2=15$.

Ans:

```
Clu1<-rnorm(100,0,1)
Clu2<-rnorm(100,2,5)
Clu3<-rnorm(100,4,9)
Clus<-data.frame(Clu1,Clu2,Clu3)
pop<-t(Clus)
#1st Stage Sampling
n1=2;n2=15;
w<-pop[sample(nrow(pop),n1),]
sc1<-w[1,]
sc2<-w[2,]
#2nd Stage Sampling
s2_sc1<-sample(sc1,n2)
s2_sc2<-sample(sc2,n2)
f_s=c(s2_sc1,s2_sc2)
est=mean(f_s)
```

Q.Find Bias of Ratio Estimator in Two Stage Sampling.

Ans:

[imp mcq's](#)

The Ratio Estimator of mean under two stage sampling when the first stage units are equal.

$$\bar{y}_{r2s} = \frac{\bar{\bar{y}}}{\bar{x}} \bar{X}$$

The estimator of mean in two stage sampling is

$$\bar{\bar{y}} = \sum_{i=1}^n \sum_{j=1}^m \frac{y_{ij}}{nm} = \sum_{i=1}^n \frac{1}{n} \bar{y}_i$$

$$V(\bar{\bar{y}}) = \frac{1}{n} \left[\frac{M-m}{Mm} S_2^2 \right] + \left(\frac{N-n}{Nn} \right) S_1^2$$

Some important notations

$$e_0 = (\bar{\bar{y}} - \bar{Y}) / \bar{Y}, e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

$$E(e_0) = E(e_1) = 0, E(e_0^2) = V_{02}, E(e_1^2) = V_{20}, E(e_0 e_1) = V_{11}$$

$$V_{02} = \frac{V(\bar{y})}{\bar{Y}^2} = \frac{1}{n\bar{Y}^2} \left[\frac{M-m}{Mm} S_{2y}^2 \right] + \left(\frac{N-n}{Nn} \right) S_{1y}^2$$

$$V_{20} = \frac{V(\bar{x})}{\bar{X}^2} = \frac{1}{n\bar{X}^2} \left[\frac{M-m}{Mm} S_{2x}^2 \right] + \left(\frac{N-n}{Nn} \right) S_{1x}^2$$

$$V_{11} = \frac{1}{\bar{Y}\bar{X}} \left\{ \frac{1}{nN} \sum_{i=1}^N \left(\frac{M-m}{Mm} \right) (\rho_{xy2i} S_{y2i} S_{x2i}) + \left(\frac{N-n}{Nn} \right) (\rho_{xy1} S_{y1} S_{x1}) \right\}$$

$$S_{y2i}^2 = \frac{1}{M_i - 1} \sum_{j=1}^{M_i} (y_{ij} - \bar{Y}_i)^2$$

$$\rho_{xy2i} = \frac{S_{xy2i}}{S_{x2i} S_{y2i}}$$

$$\bar{y} = \bar{Y}(1 + e_0), \bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_{r2s} = \frac{\bar{y}}{\bar{x}} \bar{X}, \bar{y}_{r2s} = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

Bias of Ratio Estimator Under Two Stage Sampling

$$\bar{y}_{r2s} = \bar{Y}(1 + e_0)(1 + e_1)^{-1} \Rightarrow \bar{y}_{r2s} \approx \bar{Y}(1 + e_0)(1 - e_1 + e_1^2)$$

$$\bar{y}_{r2s} \approx \bar{Y}(1 - e_1 + e_1^2 + e_0 - e_0 e_1 + e_0 e_1^2)$$

$$\bar{y}_{r2s} \approx \bar{Y} + \bar{Y}(e_0 - e_1 + e_1^2 - e_0 e_1 + e_0 e_1^2)$$

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$$\bar{y}_{r2s} - \bar{Y} \approx \bar{Y}(e_0 - e_1 + e_1^2 - e_0 e_1)$$

$$E(\bar{y}_{r2s} - \bar{Y}) \approx \bar{Y}E(e_0 - e_1 + e_1^2 - e_0 e_1)$$

$$\text{Bias}(\bar{y}_{r2s}) \approx \bar{Y}(V_{20} - V_{11})$$

Q. Find MSE of Ratio Estimator in Two Stage Sampling.

Ans:

The Ratio Estimator of mean under two stage sampling when the first stage units are equal

$$\bar{y}_{r2s} = \frac{\bar{y}}{\bar{x}} \bar{X}$$

Notations

$$E(e_0) = E(e_1) = 0, E(e_0^2) = V_{02}, E(e_1^2) = V_{20}, E(e_0 e_1) = V_{11}$$

$$e_0 = (\bar{y} - \bar{Y}) / \bar{Y}, e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

$$\bar{\bar{y}} = \bar{Y}(1 + e_0), \bar{\bar{x}} = \bar{X}(1 + e_1)$$

long question

$$\bar{y}_{r2s} = \frac{\bar{\bar{y}}}{\bar{\bar{x}}} \bar{X}$$

$$\bar{y}_{r2s} = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

MSE of Ratio Estimator Under Two Stage Sampling

$$\bar{y}_{r2s} = \bar{Y}(1 + e_0)(1 + e_1)^{-1} \Rightarrow \bar{y}_{r2s} \approx \bar{Y}(1 + e_0)(1 - e_1)$$

$$\bar{y}_{r2s} \approx \bar{Y}(1 + e_0 - e_1 - e_0 e_1) \Rightarrow \bar{y}_{r2s} \approx \bar{Y} + \bar{Y}(e_0 - e_1)$$

$$E(\bar{y}_{r2s} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0 - e_1)^2$$

$$E(\bar{y}_{r2s} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0^2 + e_1^2 - 2e_0 e_1)$$

$$MSE(\bar{y}_{r2s}) \approx \bar{Y}^2 (V_{02} + V_{20} - 2V_{11})$$

Q.Find Bias of Product Estimator in Two Stage Sampling.

Ans:

The Product Estimator of mean under two stage sampling when the first stage units are equal

$$\bar{y}_{p2s} = \frac{\bar{\bar{y}} \bar{\bar{x}}}{\bar{X}}$$

Some important notations

$$E(e_0) = E(e_1) = 0, E(e_0^2) = V_{02}$$

$$E(e_1^2) = V_{20}, E(e_0 e_1) = V_{11}$$

$$e_0 = (\bar{y} - \bar{Y}) / \bar{Y}, e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

$$\bar{\bar{y}} = \bar{Y}(1 + e_0), \bar{\bar{x}} = \bar{X}(1 + e_1)$$

$$\bar{y}_{p2s} = \frac{\bar{\bar{y}} \bar{\bar{x}}}{\bar{X}}$$

$$\bar{y}_{p2s} = \frac{\bar{Y}(1 + e_0)(1 + e_1)}{\bar{X}} \bar{X}$$

Bias of Ratio Estimator Under Two Stage Sampling

$$\bar{y}_{p2s} = \bar{Y}(1+e_0)(1+e_1) \Rightarrow \bar{y}_{p2s} = \bar{Y}(1+e_0+e_1+e_0e_1)$$

$$\bar{y}_{p2s} = \bar{Y} + \bar{Y}(e_0+e_1+e_0e_1)$$

$$E(\bar{y}_{p2s} - \bar{Y}) = \bar{Y}E(e_0+e_1+e_0e_1)$$

$$\text{Bias}(\bar{y}_{p2s}) = \bar{Y}(V_{11})$$

Q.Find MSE of Product Estimator in Two Stage Sampling

Ans:

The Product Estimator of mean under two stage sampling when the first stage units are equal

$$\bar{y}_{p2s} = \frac{\bar{\bar{y}}\bar{\bar{x}}}{\bar{X}}$$

Notations

$$E(e_0) = E(e_1) = 0,$$

$$E(e_0^2) = V_{02}, E(e_1^2) = V_{20}$$

$$E(e_0e_1) = V_{11}$$

$$e_0 = (\bar{\bar{y}} - \bar{Y})/\bar{Y}$$

$$e_1 = (\bar{\bar{x}} - \bar{X})/\bar{X}$$

$$\bar{\bar{y}} = \bar{Y}(1+e_0), \bar{\bar{x}} = \bar{X}(1+e_1)$$

$$\bar{y}_{p2s} = \frac{\bar{\bar{y}}\bar{\bar{x}}}{\bar{X}}, \bar{y}_{p2s} = \frac{\bar{Y}(1+e_0)(1+e_1)}{\bar{X}} \bar{X}$$

MSE of Product Estimator

$$\bar{y}_{p2s} = \bar{Y}(1+e_0)(1+e_1)$$

$$\bar{y}_{p2s} = \bar{Y}(1+e_0+e_1+e_0e_1)$$

$$\bar{y}_{p2s} \approx \bar{Y} + \bar{Y}(e_0+e_1)$$

$$E(\bar{y}_{p2s} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0+e_1)^2 \approx \bar{Y}^2 E(e_0^2 + e_1^2 + 2e_0e_1)$$

$$MSE(\bar{y}_{p2s}) \approx \bar{Y}^2 (V_{02} + V_{20} + 2V_{11})$$

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Ranked set sampling

mcq's

Introduction:

Ranked set sampling (RSS) is an alternative to simple random sampling that can sometimes offer large improvements in precision. McIntyre (1952) introduced the basic concept of ranked set sampling in order to estimate the population means of pasture and forage yields. The RSS procedure is elaborated as follows

First, a simple random sample of size k is drawn from the population and the k sampling units are ranked with respect to the variable of interest, say X , by judgment without actual measurement. Then the unit with rank 1 is identified and taken for the measurement of X . Next, another simple random sample of size k is drawn and the units of the sample are ranked by judgment, the unit with rank 2 is taken for the measurement of X and the remaining units are discarded. Then the unit with rank 2 is identified and taken for the measurement of X . Then another sample of size 3.

$x_{1(1)1}$	$x_{1(2)1}$	$x_{1(3)1}$	$x_{1(4)1}$	$x_{1(5)1}$	$x_{1(6)1}$
$x_{2(1)1}$	$x_{2(2)1}$	$x_{2(3)1}$	$x_{2(4)1}$	$x_{2(5)1}$	$x_{2(6)1}$
$x_{3(1)1}$	$x_{3(2)1}$	$x_{3(3)1}$	$x_{3(4)1}$	$x_{3(5)1}$	$x_{3(6)1}$
$x_{4(1)1}$	$x_{4(2)1}$	$x_{4(3)1}$	$x_{4(4)1}$	$x_{4(5)1}$	$x_{4(6)1}$
$x_{5(1)1}$	$x_{5(2)1}$	$x_{5(3)1}$	$x_{5(4)1}$	$x_{5(5)1}$	$x_{5(6)1}$
$x_{6(1)1}$	$x_{6(2)1}$	$x_{6(3)1}$	$x_{6(4)1}$	$x_{6(5)1}$	$x_{6(6)1}$

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Ranking with Auxiliary Variable

In a case when we are using the auxiliary information (or variable) for ranking the variable of interest, then ranking error depend on the correlation. The performance of the RSS estimator is actually based on the ranking of the sample units in variable of interest

Q. Prove that sample mean is an unbiased estimator of mean under ranked set sampling

Ans:

The mean Estimator in RSS is

$$\bar{X}^* = \frac{\sum_{i=1}^k X_{(i)}^*}{k}$$

The Expected Value is

$$E(\bar{X}^*) = \frac{\sum_{i=1}^k E(X_{(i)}^*)}{k}$$

We have assumed perfect rankings. $X_{(i)}^*$ is distributed like the i^{th} order statistic from a continuous distribution with p.d.f. $f(x)$ and c.d.f. $F(x)$.

$$E\left(X_{(i)}^*\right)=\int_{-\infty}^{\infty} x \frac{k!}{(i-1)(k-i)!}[F(x)]^{i-1}[1-F(x)]^{k-i} f(x) dx$$

$$E\left(\bar{X}^*\right)=\frac{\sum_{i=1}^k E\left(X_{(i)}^*\right)}{k}$$

$$E\left(\bar{X}^*\right)=\frac{1}{k} \sum_{i=1}^k \left\{ \int_{-\infty}^{\infty} kx \frac{(k-1)!}{(i-1)(k-i)!}[F(x)]^{i-1}[1-F(x)]^{k-i} f(x) dx \right\}$$

$$E\left(\bar{X}^*\right)=\frac{1}{k} \sum_{i=1}^k \left\{ \int_{-\infty}^{\infty} kx \binom{k-1}{i-1} [F(x)]^{i-1} [1-F(x)]^{k-i} f(x) dx \right\}$$

$$E\left(\bar{X}^*\right)=\int_{-\infty}^{\infty} x f(x) \sum_{i=1}^k \left[\binom{k-1}{i-1} [F(x)]^{i-1} [1-F(x)]^{k-i} \right] dx$$

$$E\left(\bar{X}^*\right)=\int_{-\infty}^{\infty} x f(x) dx = \bar{X}$$

$$E\left(\bar{X}^*\right)=\bar{X}$$

Q. Find variance expression for mean estimator under ranked set sampling.

Ans:

The mean Estimator in RSS is

$$\bar{X}^* = \frac{\sum_{i=1}^k X_{(i)}^*}{k}$$

By definition

$$E\left(X_{(i)}^* - \bar{X}\right)^2 = E\left(X_{(i)}^* - \bar{X}_{(i)}^* + \bar{X}_{(i)}^* - \bar{X}\right)^2$$

$$E\left(X_{(i)}^* - \bar{X}\right)^2 = E\left(X_{(i)}^* - \bar{X}_{(i)}^*\right)^2 + E\left(\bar{X}_{(i)}^* - \bar{X}\right)^2$$

$$E\left(X_{(i)}^* - \bar{X}_{(i)}^*\right)^2 = V\left(X_{(i)}^*\right) = E\left(X_{(i)}^* - \bar{X}\right)^2 - E\left(\bar{X}_{(i)}^* - \bar{X}\right)^2$$

$$V(\bar{X}^*) = \frac{1}{k^2} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$

Taking,

$$\begin{aligned} \sum_{i=1}^k E(X_{(i)}^* - \bar{X}_{(i)}^*)^2 &= \sum_{i=1}^k \left\{ \int_{-\infty}^{\infty} k(x - \bar{X})^2 \binom{k-1}{i-1} [F(x)]^{i-1} [1-F(x)]^{k-i} f(x) dx \right\} \\ &= k \int_{-\infty}^{\infty} (x - \bar{X})^2 f(x) \sum_{i=1}^k \left[\binom{k-1}{i-1} [F(x)]^{i-1} [1-F(x)]^{k-i} \right] dx \\ V(\bar{X}^*) &= \frac{1}{k^2} \left[k\sigma^2 - \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2 \right] \end{aligned}$$

$$V(\bar{X}^*) = \frac{\sigma^2}{k} - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$

143. Generate a population of size 1000 following normal distribution with mean 0 and variance 1. Further calculate the sample mean using ranked set sampling taking sample size as 6.

Ans:

```
k=6;rssx=matrix(k,k);
pop=rnorm(1000,0,1)
for(i in 1:k)
{
s=sample(pop,k)
xs=sort(s,decreasing = FALSE)
rssx[i,]=xs
}
rssx_s=diag(rssx)
est_srs=mean(x)
est_rss=mean(rssx_s)
rssx
```

Q. Find Bias of Ratio Estimator in Ranked Set Sampling such that

$$E(e_0) = E(e_1) = 0$$

$$E(e_1^2) = \frac{1}{rk} \frac{\sigma_x^2}{\bar{X}^2} - \frac{1}{\bar{X}^2 rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2 = \eta C_x^2 - D_{x[i]}^2$$

$$E(e_0^2) = \frac{1}{rk} \frac{\sigma_y^2}{\bar{Y}^2} - \frac{1}{\bar{Y}^2 rk^2} \sum_{i=1}^k (\bar{Y}_{(i)}^* - \bar{Y})^2 = \eta C_y^2 - D_{y[i]}^2$$

$$E(e_0 e_1) = \frac{1}{rk} \frac{\sigma_{yx}}{\bar{Y}\bar{X}} - \frac{1}{\bar{Y}\bar{X} rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})(\bar{Y}_{(i)}^* - \bar{Y}) = \eta C_{xy} - D_{xy[i]}^2$$

Ans:

The ratio estimator of mean is

$$\bar{y}_{[rss]} = \frac{\bar{Y}^*}{\bar{X}^*} \bar{X}$$

The Variance of Sample Mean Under RSS

$$V(\bar{X}^*) = \frac{\sigma_x^2}{k} - \frac{1}{k^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2$$

Notations

$$e_0 = (\bar{Y}^* - \bar{Y}) / \bar{Y}, e_1 = (\bar{X}^* - \bar{X}) / \bar{X}$$

$$E(e_0) = E(e_1) = 0$$

$$E(e_1^2) = \frac{1}{rk} \frac{\sigma_x^2}{\bar{X}^2} - \frac{1}{\bar{X}^2 rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2 = \eta C_x^2 - D_{x[i]}^2$$

$$E(e_0^2) = \frac{1}{rk} \frac{\sigma_y^2}{\bar{Y}^2} - \frac{1}{\bar{Y}^2 rk^2} \sum_{i=1}^k (\bar{Y}_{(i)}^* - \bar{Y})^2 = \eta C_y^2 - D_{y[i]}^2$$

$$E(e_0 e_1) = \frac{1}{rk} \frac{\sigma_{yx}}{\bar{Y}\bar{X}} - \frac{1}{\bar{Y}\bar{X} rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})(\bar{Y}_{(i)}^* - \bar{Y}) = \eta C_{xy} - D_{xy[i]}^2$$

$$\bar{Y}^* = \bar{Y}(1 + e_0)$$

$$\bar{X}^* = \bar{X}(1 + e_1)$$

$$\bar{y}_{[rss]} = \frac{\bar{Y}^*}{\bar{X}^*} \bar{X}$$

$$\bar{y}_{[rss]} = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

Bias of Ratio Estimator Under RS Sampling

$$\bar{y}_{[rss]} = \bar{Y}(1+e_0)(1+e_1)^{-1} \Rightarrow \bar{y}_{[rss]} \approx \bar{Y}(1+e_0)(1-e_1+e_1^2)$$

$$\bar{y}_{[rss]} \approx \bar{Y}(1-e_1+e_1^2+e_0-e_0e_1+e_0e_1^2)$$

$$\bar{y}_{[rss]} \approx \bar{Y} + \bar{Y}(e_0 - e_1 + e_1^2 - e_0e_1 + e_0e_1^2)$$

$$\bar{y}_{[rss]} - \bar{Y} \approx \bar{Y}(e_0 - e_1 + e_1^2 - e_0e_1)$$

$$E(\bar{y}_{[rss]} - \bar{Y}) \approx \bar{Y}E(e_0 - e_1 + e_1^2 - e_0e_1)$$

$$Bias(\bar{y}_{[rss]}) \approx \bar{Y} \begin{bmatrix} \eta C_x^2 - D_{x[i]}^2 \\ -(\eta C_{xy} - D_{xy[i]}) \end{bmatrix}$$

Q. Find MSE of Ratio Estimator in Ranked Set Sampling such that

$$E(e_0) = E(e_1) = 0$$

$$E(e_1^2) = \frac{1}{rk} \frac{\sigma_x^2}{\bar{X}^2} - \frac{1}{\bar{X}^2 rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2 = \eta C_x^2 - D_{x[i]}^2$$

$$E(e_0^2) = \frac{1}{rk} \frac{\sigma_y^2}{\bar{Y}^2} - \frac{1}{\bar{Y}^2 rk^2} \sum_{i=1}^k (\bar{Y}_{(i)}^* - \bar{Y})^2 = \eta C_y^2 - D_{y[i]}^2$$

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$$E(e_0e_1) = \frac{1}{rk} \frac{\sigma_{yx}}{\bar{Y}\bar{X}} - \frac{1}{\bar{Y}\bar{X}rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})(\bar{Y}_{(i)}^* - \bar{Y}) = \eta C_{xy} - D_{xy[i]}$$

Ans:

The Ratio Estimator of mean for ranked set sampling is

$$\bar{y}_{[rss]} = \frac{\bar{Y}^*}{\bar{X}^*} \bar{X}$$

Notations

$$e_0 = (\bar{Y}^* - \bar{Y})/\bar{Y}, e_1 = (\bar{X}^* - \bar{X})/\bar{X}$$

$$E(e_0) = E(e_1) = 0,$$

$$E(e_1^2) = \frac{1}{rk} \frac{\sigma_x^2}{\bar{X}^2} - \frac{1}{\bar{X}^2 rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})^2 = \eta C_x^2 - D_{x[i]}^2$$

$$E(e_0^2) = \frac{1}{rk} \frac{\sigma_y^2}{\bar{Y}^2} - \frac{1}{\bar{Y}^2 rk^2} \sum_{i=1}^k (\bar{Y}_{(i)}^* - \bar{Y})^2 = \eta C_y^2 - D_{y[i]}^2$$

$$E(e_0 e_1) = \frac{1}{rk} \frac{\sigma_{yx}}{\bar{Y}\bar{X}} - \frac{1}{\bar{Y}\bar{X}rk^2} \sum_{i=1}^k (\bar{X}_{(i)}^* - \bar{X})(\bar{Y}_{(i)}^* - \bar{Y}) = \eta C_{xy} - D_{xy[i]}$$

$$\bar{Y}^* = \bar{Y}(1 + e_0)$$

$$\bar{X}^* = \bar{X}(1 + e_1)$$

$$\bar{y}_{[rss]} = \frac{\bar{Y}^*}{\bar{X}^*} \bar{X}$$

$$\bar{y}_{[rss]} = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

MSE of Ratio Estimator Under RS Sampling

$$\bar{y}_{[rss]} = \bar{Y}(1 + e_0)(1 + e_1)^{-1} \Rightarrow \bar{y}_{[rss]} \approx \bar{Y}(1 + e_0)(1 - e_1)$$

$$\bar{y}_{[rss]} \approx \bar{Y}(1 - e_1 + e_0) \Rightarrow \bar{y}_{[rss]} \approx \bar{Y} + \bar{Y}(e_0 - e_1)$$

$$E(\bar{y}_{[rss]} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0 - e_1)^2$$

$$MSE(\bar{y}_{[rss]}) \approx \bar{Y}^2 E(e_0^2 + e_1^2 - 2e_0 e_1)$$

$$MSE(\bar{y}_{[rss]}) \approx \bar{Y}^2 (\eta C_y^2 - D_{y[i]}^2 + \eta C_x^2 - D_{x[i]}^2 - 2(\eta C_{xy} - D_{xy[i]}))$$

$$MSE(\bar{y}_{[rss]}) \approx \bar{Y}^2 \begin{bmatrix} \eta(C_y^2 + C_x^2 - 2C_{xy}) \\ -D_{y[i]}^2 - D_{x[i]}^2 + 2D_{xy[i]} \end{bmatrix}$$

Similarly, the Bias and MSE of Product Estimator Under RSS

The Product Estimator of mean is

$$\bar{y}_{[pss]} = \frac{\bar{Y}^*}{\bar{X}} \bar{X}^*$$

Bias of Product estimator is

$$Bias(\bar{y}_{[pss]}) \approx \bar{Y} [\eta C_{xy} - D_{xy[i]}]$$

MSE of product estimator

$$MSE(\bar{y}_{[pss]}) \approx \bar{Y}^2 \left[\eta(C_y^2 + C_x^2 + 2C_{xy}) - D_{y[i]}^2 - D_{x[i]}^2 - 2D_{xy[i]} \right]$$

Q. Write some types of sampling schemes using ranked set sampling.

Ranked Set Sampling

$x_{1(1)1}$	$x_{1(2)1}$	$x_{1(3)1}$	$x_{1(4)1}$	$x_{1(5)1}$	$x_{1(6)1}$
$x_{2(1)1}$	$x_{2(2)1}$	$x_{2(3)1}$	$x_{2(4)1}$	$x_{2(5)1}$	$x_{2(6)1}$
$x_{3(1)1}$	$x_{3(2)1}$	$x_{3(3)1}$	$x_{3(4)1}$	$x_{3(5)1}$	$x_{3(6)1}$
$x_{4(1)1}$	$x_{4(2)1}$	$x_{4(3)1}$	$x_{4(4)1}$	$x_{4(5)1}$	$x_{4(6)1}$
$x_{5(1)1}$	$x_{5(2)1}$	$x_{5(3)1}$	$x_{5(4)1}$	$x_{5(5)1}$	$x_{5(6)1}$
$x_{6(1)1}$	$x_{6(2)1}$	$x_{6(3)1}$	$x_{6(4)1}$	$x_{6(5)1}$	$x_{6(6)1}$

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Extreme Ranked Set Sampling

Samawi et al. (1996) suggested a modified RSS design named as Extreme ranked set sampling (ERSS).

$x_{1(1)1}$	$x_{1(2)1}$	$x_{1(3)1}$	$x_{1(4)1}$	$x_{1(5)1}$	$x_{1(6)1}$
$x_{2(1)1}$	$x_{2(2)1}$	$x_{2(3)1}$	$x_{2(4)1}$	$x_{2(5)1}$	$x_{2(6)1}$
$x_{3(1)1}$	$x_{3(2)1}$	$x_{3(3)1}$	$x_{3(4)1}$	$x_{3(5)1}$	$x_{3(6)1}$
$x_{4(1)1}$	$x_{4(2)1}$	$x_{4(3)1}$	$x_{4(4)1}$	$x_{4(5)1}$	$x_{4(6)1}$
$x_{5(1)1}$	$x_{5(2)1}$	$x_{5(3)1}$	$x_{5(4)1}$	$x_{5(5)1}$	$x_{5(6)1}$
$x_{6(1)1}$	$x_{6(2)1}$	$x_{6(3)1}$	$x_{6(4)1}$	$x_{6(5)1}$	$x_{6(6)1}$

Median Ranked set sampling

$X_{1(1)1}$	$X_{1(2)1}$	$X_{1(3)1}$	$X_{1(4)1}$	$X_{1(5)1}$	$X_{1(6)1}$
$X_{2(1)1}$	$X_{2(2)1}$	$X_{2(3)1}$	$X_{2(4)1}$	$X_{2(5)1}$	$X_{2(6)1}$
$X_{3(1)1}$	$X_{3(2)1}$	$X_{3(3)1}$	$X_{3(4)1}$	$X_{3(5)1}$	$X_{3(6)1}$
$X_{4(1)1}$	$X_{4(2)1}$	$X_{4(3)1}$	$X_{4(4)1}$	$X_{4(5)1}$	$X_{4(6)1}$
$X_{5(1)1}$	$X_{5(2)1}$	$X_{5(3)1}$	$X_{5(4)1}$	$X_{5(5)1}$	$X_{5(6)1}$
$X_{6(1)1}$	$X_{6(2)1}$	$X_{6(3)1}$	$X_{6(4)1}$	$X_{6(5)1}$	$X_{6(6)1}$

$X_{1(1)1}$	$X_{1(2)1}$	$X_{1(3)1}$	$X_{1(4)1}$	$X_{1(5)1}$
$X_{2(1)1}$	$X_{2(2)1}$	$X_{2(3)1}$	$X_{2(4)1}$	$X_{2(5)1}$

$X_{3(1)1}$	$X_{3(2)1}$	$X_{3(3)1}$	$X_{3(4)1}$	$X_{3(5)1}$
$X_{4(1)1}$	$X_{4(2)1}$	$X_{4(3)1}$	$X_{4(4)1}$	$X_{4(5)1}$
$X_{5(1)1}$	$X_{5(2)1}$	$X_{5(3)1}$	$X_{5(4)1}$	$X_{5(5)1}$

Mean in Ranked Set Sampling (RSS)

The mean Estimator in RSS is $\bar{X}^* = \frac{\sum_{i=1}^k X_{(i)}^*}{k}$

In case of r cycles $\bar{X}^* = \frac{\sum_{c=1}^r \sum_{i=1}^k X_{(i)c}^*}{rk}$

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Mean in Extreme Ranked Set Sampling (ERSS)

$$\bar{X}_{(ERSS)e}^* = \frac{1}{rk} \sum_{c=1}^r \left[\sum_{i=1}^{k/2} X_{i,(1),c}^* + \sum_{i=1}^{k/2} X_{\frac{k}{2}+i,(k),c}^* \right]$$

$$\bar{X}_{(ERSS)o}^* = \frac{1}{rk} \sum_{c=1}^r \left[\sum_{i=1}^{(k-1)/2} X_{i,(1),c}^* + \sum_{i=1}^{(k-1)/2} X_{\frac{k-1}{2}+i,(k),c}^* + X_{k,\left(\frac{k+1}{2}\right),c}^* \right]$$

Mean in Median Ranked Set Sampling (MRSS)

The mean Estimator in Case of even set size

$$\bar{X}_{(MRSS)e}^* = \frac{1}{rk} \sum_{c=1}^r \left[\sum_{i=1}^{k/2} X_{i,\left(\frac{k}{2}\right),c}^* + \sum_{i=1}^{k/2} X_{\left(\frac{k}{2}+i\right),\frac{k+2}{2},c}^* \right]$$

The mean Estimator in Case of odd set size

$$\bar{X}_{(MRSS)o}^* = \frac{1}{rk} \sum_{c=1}^r \left[\sum_{i=1}^k X_{i,\left(\frac{k+1}{2}\right),c}^* \right]$$

Pair Ranked Set Sampling and Double Ranked Set Sampling are further developments in Ranked set sampling.

Dealing with Non-response

Introduction:

The individuals chosen for the sample are not ready to participate in the survey. This is a type of selection bias Unit/Item nonresponse. The problem of non-response can be dealt with using following methods.

Sub-sampling of non-respondents

Randomized response technique.

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Hansen and Hurwitz (1946) technique

Taking a sub sample of non-respondents after the first mail attempt and then enumerating the sub sample of non-respondents by personal interviews

Hansen & Hurwitz (1946) Estimator $\bar{y} = \frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2$

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Q. Prove that following Hansen and Hurwitz estimator is unbiased to population mean.

$$\bar{y} = \frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2$$

Ans:

Hansen & Hurwitz (1946) Estimator

$$\bar{y}^* = \frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2'$$

$$E(\bar{y}^*) = E\left(\frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2'\right)$$

$$E(\bar{y}^*) = E_1 E_2 \left(\frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2'\right)$$

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Hansen & Hurwitz (1946) is Unbiased

$$E(\bar{y}^*) = \left(\frac{n_1}{n} \bar{Y} + \frac{n_2}{n} \bar{Y}\right)$$

$$E(\bar{y}^*) = \bar{Y} \left(\frac{n_1}{n} + \frac{n_2}{n}\right)$$

$$E(\bar{y}^*) = \bar{Y}$$

must paper

Q. Find Variance of following Hansen & Hurwitz (1946) Estimator.

$$\bar{y} = \frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2'$$

Ans:

The Hansen & Hurwitz (1946) Estimator is

$$\bar{y}^* = \frac{n_1}{n} \bar{y}_1 + \frac{n_2}{n} \bar{y}_2'$$

Variance

$$\bar{y}^* = \frac{1}{n} (n_1 \bar{y}_1 + n_2 \bar{y}_2) + \frac{n_2}{n} (\bar{y}_2' - \bar{y}_2)$$

$$V(\bar{y}^*) = V\left[\frac{1}{n} (n_1 \bar{y}_1 + n_2 \bar{y}_2) + \frac{n_2}{n} (\bar{y}_2' - \bar{y}_2)\right]$$

$$V\left[\frac{1}{n}(n_1\bar{y}_1 + n_2\bar{y}_2)\right] = \theta S_y^2$$

$$\frac{n_2^2}{n^2} V(\bar{y}_2' - \bar{y}_2) = \frac{n_2^2}{n^2} E(\bar{y}_2' - \bar{y}_2)^2$$

$$\frac{n_2^2}{n^2} E(\bar{y}_2' - \bar{y}_2)^2 = \frac{n_2^2}{n^2} E\left[(\bar{y}_2' - \bar{Y}_2) - (\bar{y}_2 - \bar{Y}_2)\right]^2$$

$$= \frac{n_2^2}{n^2} \left[E(\bar{y}_2' - \bar{Y}_2)^2 + E(\bar{y}_2 - \bar{Y}_2)^2 - 2E(\bar{y}_2' - \bar{Y}_2)(\bar{y}_2 - \bar{Y}_2) \right]$$

$$= \frac{n_2^2}{n^2} \left[E(\bar{y}_2' - \bar{Y}_2)^2 - E(\bar{y}_2 - \bar{Y}_2)^2 \right]$$

$$= \frac{n_2^2}{n^2} \left(\frac{1}{\frac{n_2}{k}} - \frac{1}{n_2} \right) S_{y(2)}^2$$

$$= \frac{n_2}{n^2} (k-1) S_{y(2)}^2$$

$$= \lambda S_{y(2)}^2$$

$$V(\bar{y}^*) = \theta S_y^2 + \lambda S_{y(2)}^2 \quad \text{imp mcq's}$$

150. Generate a population consisting on 1000 values in R program and calculate Hansen & Hurwitz (1946) Estimator such that $n_1=80; n_2=20; r=10$.

```
N=1000; n=100; n1=80; n2=20; r=10;
pop <- rnorm(N,0,1)
s <- sample(pop,n)
s_r <- pop[1:80]
s_nr <- pop[81:100]
s2 <- sample(s_nr,r)
```

imp s/q

```
m <- (n1/n)*mean(s_r) + (n2/n)*mean(s2)
```

Q. Find the expression of Bias of ratio estimator in case of nonresponse on study variable only, where

$$E(e_0^*) = E(e_1) = 0,$$

$$E(e_1^2) = \left(\frac{N-n}{nN} \right) \frac{\sigma_x^2}{\bar{X}^2} = \theta C_x^2$$

$$E(e_0^{*2}) = \frac{1}{\bar{Y}^2} [\theta S_y^2 + \lambda S_{y(2)}^2] = C_y^{*2}$$

$$E(e_0^* e_1) = \theta \frac{S_{yx}}{\bar{Y}\bar{X}} = \theta C_{yx}$$

Ans:

Ratio estimator when non-response occurs in study variable y is as follow

$$\bar{y}_r^* = \frac{\bar{y}^*}{\bar{x}} \bar{X}$$

$$V(\bar{y}^*) = \theta S_y^2 + \lambda S_{y(2)}^2$$

Notations

$$e_0^* = (\bar{y}^* - \bar{Y}) / \bar{Y}, e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

$$E(e_0^*) = E(e_1) = 0,$$

$$E(e_1^2) = \left(\frac{N-n}{nN} \right) \frac{\sigma_x^2}{\bar{X}^2} = \theta C_x^2$$

$$E(e_0^{*2}) = \frac{1}{\bar{Y}^2} [\theta S_y^2 + \lambda S_{y(2)}^2] = C_y^{*2}$$

$$E(e_0^* e_1) = \theta \frac{S_{yx}}{\bar{Y}\bar{X}} = \theta C_{yx}$$

$$\bar{y}^* = \bar{Y}(1 + e_0^*)$$

$$\bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_r^* = \frac{\bar{y}^*}{\bar{x}} \bar{X} \longrightarrow \bar{y}_r^* = \frac{\bar{Y}(1 + e_0^*)}{\bar{X}(1 + e_1)} \bar{X}$$

Bias of Ratio Estimator Under Non-response

$$\bar{y}_r^* = \bar{Y}(1+e_0^*)(1+e_1)^{-1} \Rightarrow \bar{y}_r^* \approx \bar{Y}(1+e_0^*)(1-e_1+e_1^2)$$

$$\bar{y}_r^* \approx \bar{Y}(1-e_1+e_1^2+e_0^*-e_0^*e_1+e_0^*e_1^2)$$

$$\bar{y}_r^* - \bar{Y} \approx \bar{Y}(e_0^* - e_1 + e_1^2 - e_0^*e_1 + e_0^*e_1^2)$$

$$\bar{y}_r^* - \bar{Y} \approx \bar{Y}(e_0^* - e_1 + e_1^2 - e_0^*e_1)$$

$$E(\bar{y}_r^* - \bar{Y}) \approx \bar{Y}E(e_0^* - e_1 + e_1^2 - e_0^*e_1)$$

$$\text{Bias}(\bar{y}_r^*) \approx \theta \bar{Y} [C_x^2 - C_{yx}]$$

Q. Find the expression of MSE of ratio estimator in case of nonresponse on study variable only, where

$$E(e_0^*) = E(e_1) = 0,$$

$$E(e_1^2) = \left(\frac{N-n}{nN} \right) \frac{\sigma_x^2}{\bar{X}^2} = \theta C_x^2$$

$$E(e_0^{*2}) = \frac{1}{\bar{Y}^2} [\theta S_y^2 + \lambda S_{y(2)}^2] = C_y^{*2}$$

$$E(e_0^*e_1) = \theta \frac{S_{yx}}{\bar{Y}\bar{X}} = \theta C_{yx}$$

Ans.

$$\bar{y}_r^* = \frac{\bar{y}^*}{\bar{x}}$$

$$\bar{y}^* = \bar{Y}(1+e_0^*)$$

$$\bar{x} = \bar{X}(1+e_1)$$

$$\bar{y}_r^* = \frac{\bar{y}^*}{\bar{x}}$$

$$\bar{y}_r^* = \frac{\bar{Y}(1+e_0^*)}{\bar{X}(1+e_1)} \bar{X}$$

$$\bar{y}_r^* = \bar{Y}(1+e_0^*)(1+e_1)^{-1} \Rightarrow \bar{y}_r^* \approx \bar{Y}(1+e_0^*)(1-e_1)$$

$$\bar{y}_r^* - \bar{Y} \approx \bar{Y}(e_0^* - e_1)$$

$$E(\bar{y}_r^* - \bar{Y})^2 \approx \bar{Y}^2 E(e_0^* - e_1)^2$$

$$MSE(\bar{y}_r^*) \approx \bar{Y}^2 E(e_0^{*2} + e_1^2 - 2e_0^*e_1)$$

$$MSE(\bar{y}_r^*) \approx \bar{Y}^2 (C_y^{*2} + \theta C_x^2 - 2\theta C_{yx})$$

Similary, Bias and MSE of Product Estimator

Product estimator when non-response occurs in study variable y is follow as

$$\bar{y}_p^* = \frac{\bar{y}^* \bar{x}}{\bar{X}}$$

Bias and MSE of Product Estimator Under Non-response

$$Bias(\bar{y}_p^*) = \theta \bar{Y} [C_{yx}]$$

$$MSE(\bar{y}_p^*) \approx \bar{Y}^2 (C_y^{*2} + \theta C_x^2 + 2\theta C_{yx})$$

Q. Find Bias and MSE of Ratio Estimator in Case of Non-response on Both Variable, where

$$E(e_0^*) = E(e_1^*) = 0,$$

$$E(e_0^{*2}) = \frac{1}{\bar{Y}^2} [\theta S_y^2 + \lambda S_{y(2)}^2] = C_{y'}^{*2}$$

$$E(e_1^{*2}) = \frac{1}{\bar{X}^2} [\theta S_x^2 + \lambda S_{x(2)}^2] = C_x^{*2}$$

$$E(e_0^* e_1^*) = \frac{1}{\bar{Y}\bar{X}} [\theta S_{xy} + \lambda S_{xy(2)}^2] = C_{xy}^*$$

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Ans

Ratio estimator when non-response occurs in study variable y is follow as

$$\bar{y}_r^* = \frac{\bar{y}^*}{\bar{x}^*} \bar{X}$$

Variance of Hansen & Hurwitz (1946) Estimator

$$V(\bar{y}^*) = \theta S_y^2 + \lambda S_{y(2)}^2$$

$$V(\bar{x}^*) = \theta S_x^2 + \lambda S_{x(2)}^2$$

$$e_0^* = (\bar{y}^* - \bar{Y}) / \bar{Y}$$

$$e_1^* = (\bar{x}^* - \bar{X}) / \bar{X}$$

$$E(e_0^*) = E(e_1^*) = 0,$$

$$E(e_0^{*2}) = \frac{1}{\bar{Y}^2} [\theta S_y^2 + \lambda S_{y(2)}^2] = C_y^{*2}$$

$$E(e_1^{*2}) = \frac{1}{\bar{X}^2} [\theta S_x^2 + \lambda S_{x(2)}^2] = C_x^{*2}$$

$$E(e_0^* e_1^*) = \frac{1}{\bar{Y}\bar{X}} [\theta S_{xy} + \lambda S_{xy(2)}^2] = C_{xy}^*$$

$$\bar{y}^* = \bar{Y}(1 + e_0^*)$$

$$\bar{x}^* = \bar{X}(1 + e_1^*)$$

$$\bar{y}_r^{**} = \frac{\bar{y}^*}{\bar{x}^*} \bar{X}$$

$$\bar{y}_r^{**} = \frac{\bar{Y}(1 + e_0^*)}{\bar{X}(1 + e_1^*)} \bar{X}$$

Bias of Ratio Estimator Under Non-response

$$\bar{y}_r^{**} = \bar{Y}(1 + e_0^*)(1 + e_1^*)^{-1} \Rightarrow \bar{y}_r^{**} \approx \bar{Y}(1 + e_0^*)(1 - e_1^* + e_1^{*2})$$

$$\bar{y}_r^{**} \approx \bar{Y}(1 - e_1^* + e_1^{*2} + e_0^* - e_0^* e_1^* + e_0^* e_1^{*2})$$

$$\bar{y}_r^{**} - \bar{Y} \approx \bar{Y}(e_0^* - e_1^* + e_1^{*2} - e_0^* e_1^* + e_0^* e_1^{*2})$$

$$\bar{y}_r^{**} - \bar{Y} \approx \bar{Y}(e_0^* - e_1^* + e_1^{*2} - e_0^* e_1^*)$$

$$E(\bar{y}_r^{**} - \bar{Y}) \approx \bar{Y}E(e_0^* - e_1^* + e_1^{*2} - e_0^* e_1^*)$$

$$\text{Bias}(\bar{y}_r^{**}) \approx \theta \bar{Y} [C_x^{*2} - C_{yx}^*]$$

Notations

$$\bar{y}^* = \bar{Y}(1 + e_0^*)$$

$$\bar{x}^* = \bar{X}(1 + e_1^*)$$

$$\bar{y}_r^{**} = \frac{\bar{y}^*}{\bar{x}^*} \bar{X}$$

$$\bar{y}_r^* = \frac{\bar{Y}(1 + e_0^*)}{\bar{X}(1 + e_1^*)} \bar{X}$$

MSE of Ratio Estimator Under RS Sampling

$$\bar{y}_r^{**} = \bar{Y}(1 + e_0^*)(1 + e_1^*)^{-1} \Rightarrow \bar{y}_r^{**} \approx \bar{Y}(1 + e_0^*)(1 - e_1^*)$$

$$\bar{y}_r^{**} \approx \bar{Y}(1 - e_1^* + e_0^* - e_0^* e_1^*) \Rightarrow \bar{y}_r^{**} - \bar{Y} \approx \bar{Y}(e_0^* - e_1^*)$$

$$E(\bar{y}_r^{**} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0^* - e_1^*)^2$$

$$MSE(\bar{y}_r^{**}) \approx \bar{Y}^2 E(e_0^{*2} + e_1^{*2} - 2e_0^* e_1^*)$$

$$MSE(\bar{y}_r^{**}) \approx \bar{Y}^2 (C_y^{*2} + C_x^{*2} - 2C_{yx}^*)$$

Product estimator when non-response occurs in study variable y is follow as

$$\bar{y}_p^{**} = \frac{\bar{y}^* \bar{x}^*}{\bar{X}}$$

Bias and ME of Product Estimator Under Non-response

$$Bias(\bar{y}_p^{**}) = \bar{Y} [C_{yx}^*]$$

$$MSE(\bar{y}_p^{**}) \approx \bar{Y}^2 (C_y^{*2} + C_x^{*2} + 2C_{yx}^*)$$

Randomized Response Technique

imp

Introduction:

This technique is useful to estimate the sensitive characteristics in the population. It was first proposed by S. L. Warner in 1965. Qualitative and quantitative response models. Qualitative response models are used to estimate the proportion of some behavior or occurrence in a population.

For example, to estimate the "proportion of people who smoke cigarette today"

The Warner introduced the following model

$$Z = Yp + (1 - p)(1 - Y)$$

p be the probability to answer the sensitive question and Y the true proportion of those interviewed bearing the sensitive property. Z is the proportion of YES answers.

The Warner model

$$Z = Yp + (1 - p)(1 - Y)$$

It can be transformed as

$$Y = \frac{Z + p - 1}{2p - 1}$$

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For Example

Statement 1: "I smoke cigarettes."

Statement 2: "I never smoke cigarettes."

$$Y = \frac{\frac{3}{4} + \frac{1}{6} - 1}{2\left(\frac{1}{6}\right) - 1} = \frac{1}{8}$$

Example 1.

Statement 1: "I have falsified my tax return."

Statement 2: "I have never falsified my tax return."

The Warner model

$$Z = Yp + (1 - p)(1 - Y)$$

$$\frac{40}{50} = Y\left(\frac{1}{6}\right) + \left(1 - \frac{1}{6}\right)(1 - Y)$$

$$Y = \frac{\frac{4}{5} + \frac{1}{6} - 1}{2\left(\frac{1}{6}\right) - 1}$$

Example 2.

Statement 1: "Have you ever used a sick day leave when you weren't really sick?"

Statement 2: "Have you never used a sick day leave when you weren't really sick?"

The Warner model

$$Z = Yp + (1 - p)(1 - Y)$$

$$\frac{350}{500} = Y\left(\frac{1}{6}\right) + \left(1 - \frac{1}{6}\right)(1 - Y)$$

$$Y = \frac{\frac{7}{10} + \frac{1}{6} - 1}{2\left(\frac{1}{6}\right) - 1}$$

$$Y = 0.2$$

Quantitative Randomized Response Technique.

Y may be the monthly income of the head of a household. Y may be the total value of purchase orders in a year for a company. Y may be the number of times you bunk the class during a semester.

Let S be a scrambling variable independent of Y. The respondent is asked to report a scrambled response for Y given by

$$Z = Y + S$$

Mean of Response S variable is introduced such that

$$E(S) = 0$$

$$E(Z) = E(Y)$$

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Variance of Response

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$$V(Z) = V(Y) + V(S)$$

Q. Generate a population of size 1000 using normal distribution with mean 10 and standard deviation 2. Further consider sample size n=100. Select 10,000 random samples using the model $Z=S+Y$. The variable S is taken to be normal variate with mean equal to zero and standard deviation equal 1.

- (i) Calculate the following estimator considering SRSWOR

$$t = \bar{z}$$

- (ii) Calculate the variance for above estimators

Ans:

Let S be a scrambling variable independent of Y. The respondent is asked to report a scrambled response for Y given by

$$Z = Y + S$$

Estimation of Mean

N=1000; n=100;

y <- rnorm(N, 10, 2)


```
s <- rnorm(N, 0, 1)
sa <- sample(1:N, n)
sy<-y[sa]; ss=s[sa];
z<-sy+ss
mz=mean(z)
N=1000; n=100;mz=c();
y <- rnorm(N, 10, 2)
s <- rnorm(N, 0, 1)
for(i in 1:10000)
{
sa <- sample(1:N, n)
sy<-y[sa]; ss=s[sa];
mz[i]=mean(sy+ss);
}
mean(mz)
```

Quantitative RRT with Non-Sensitive Auxiliary Variable.

Y may be the monthly income of the head of a household and X may be her current age. Y may be the total value of purchase orders in a year for a company and X may be the total turnover for that company in that year.

Scrambled Response

Let Y be the study variable, a sensitive variable which cannot be observed directly due to respondent bias. Let X be a non sensitive auxiliary variable which has a positive correlation with Y. Let S be a scrambling variable independent of Y and X. The respondent is asked to report a scrambled response for Y given by

$$Z = Y + S$$

Respondent is asked to provide a true response for X.

Mean & Variance of Z

S variable is introduced such that

$$E(S) = 0$$

$$E(Z) = E(Y)$$

$$V(Z) = V(Y) + V(S)$$

$$C_z^2 = C_y^2 + \frac{V(S)}{\bar{Y}^2}$$

imp mcq's

Ratio Estimator of Mean for Sensitive Variable

$$\bar{y}_{rs} = \frac{\bar{z}}{\bar{x}} \bar{X}$$

Q. Find Bias & MSE of Ratio Estimator Using RRT.

The Ratio Estimator is

$$\bar{y}_{rrt} = \frac{\bar{z}}{\bar{x}} \bar{X}$$

Notations

$$e_0 = (\bar{z} - \bar{Z}) / \bar{Z}, e_1 = (\bar{x} - \bar{X}) / \bar{X}$$

$$E(e_0^2) = \lambda C_z^2, E(e_1^2) = \lambda C_x^2, E(e_0 e_1) = \lambda C_{zx},$$

$$\text{where, } C_{zx} = \rho_{zx} C_z C_x$$

$$\bar{z} = \bar{Z}(1 + e_0), \bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_{rrt} = \frac{\bar{z}}{\bar{x}} \bar{X}$$

$$\bar{y}_{rrt} = \frac{\bar{Z}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

Bias of Ratio Estimator Using RRT

$$\text{Bias of Ratio Estimator} = \bar{Y}(1 + e_0)(1 + e_1)^{-1} \approx \bar{Y}(1 + e_0)(1 - e_1 + e_1^2)$$

$$\bar{y}_{rrt} = \bar{Y}(1 + e_0)(1 + e_1)^{-1}$$

$$\text{Bias}(\bar{y}_r) \approx \frac{1-f}{n} \bar{Y} (C_x^2 - \rho_{zx} C_z C_x)$$

For MSE of Ratio Estimator

$$\bar{z} = \bar{Z}(1 + e_0)$$

$$\bar{x} = \bar{X}(1 + e_1)$$

$$\bar{y}_{rrt} = \frac{\bar{z}}{\bar{x}} \bar{X}$$

$$\bar{y}_{rrt} = \frac{\bar{Z}(1 + e_0)}{\bar{X}(1 + e_1)} \bar{X}$$

$$\text{MSE of Ratio Estimator} = \bar{Y}(1 + e_0)(1 + e_1)^{-1} \approx \bar{Y}(1 + e_0)(1 - e_1)$$

$$\bar{y}_{rrt} - \bar{Y} \approx \bar{Y}(e_0 - e_1)$$

$$E(\bar{y}_{rrt} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0^2 + e_1^2 - 2e_0 e_1)$$

$$\text{MSE}(\bar{y}_{rrt}) \approx \bar{Y}^2 \lambda (C_z^2 + C_x^2 - 2\rho_{xz} C_z C_x)$$

Q. Find MSE of Ratio Estimator with Two Auxiliary Variables

ANs;

$$\bar{y}_{rr} = \bar{y} \frac{\bar{X}}{\bar{x}} \frac{\bar{Z}}{\bar{z}}$$

Notations

imp long question

$$e_0 = (\bar{y} - \bar{Y})/\bar{Y}, e_1 = (\bar{x} - \bar{X})/\bar{X}, e_2 = (\bar{z} - \bar{Z})/\bar{Z}$$

$$E(e_0) = E(e_1) = E(e_2) = 0$$

$$E(e_0^2) = \lambda C_y^2, E(e_1^2) = \lambda C_x^2, E(e_2^2) = \lambda C_z^2, E(e_0 e_1) = \lambda C_{yx},$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

$$\bar{y} = \bar{Y}(1 + e_0), \bar{x} = \bar{X}(1 + e_1), \bar{z} = \bar{Z}(1 + e_2)$$

$$\bar{y}_{rr} = \bar{y} \frac{\bar{X}}{\bar{x}} \frac{\bar{Z}}{\bar{z}}$$

$$\bar{y}_{rr} = \frac{\bar{Y}(1 + e_0)}{\bar{X}(1 + e_1)\bar{Z}(1 + e_2)} \bar{X}\bar{Z}$$

must paper may aye gya

MSE of the Estimator

$$\bar{y}_{rr} = \bar{Y}(1 + e_0)(1 + e_1)^{-1}(1 + e_2)^{-1} \approx \bar{Y}(1 + e_0)(1 - e_1)(1 - e_2)$$

$$\bar{y}_{rr} \approx \bar{Y}(1 + e_0)(1 - e_1 - e_2 + e_1 e_2) \approx \bar{Y}(1 + e_0)(1 - e_1 - e_2)$$

$$\bar{y}_{rr} \approx \bar{Y}(1 + e_0 - e_1 - e_2) \Rightarrow \bar{y}_{rr} \approx \bar{Y} + \bar{Y}(e_0 - e_1 - e_2)$$

$$\bar{y}_{rr} - \bar{Y} \approx \bar{Y}(e_0 - e_1 - e_2) \Rightarrow E(\bar{y}_{rr} - \bar{Y})^2 \approx \bar{Y}^2 E(e_0 - e_1 - e_2)^2$$

$$MSE(\bar{y}_{rr}) \approx \bar{Y}^2 E(e_0^2 + e_1^2 + e_2^2 - 2e_0 e_1 - 2e_0 e_2 + 2e_1 e_2)$$

$$MSE(\bar{y}_{rr}) \approx \lambda \bar{Y} (C_y^2 + C_x^2 + C_z^2 - 2C_{xy} - 2C_{yz} + 2C_{xz})$$

$$MSE(\bar{y}_{rr}) \approx \lambda \bar{Y} (C_y^2 + C_x^2 + C_z^2 - 2(C_{xy} + C_{yz} - C_{xz}))$$

Q. Define Regression Estimator with Two Auxiliary Variables, also find variance expression of the estimator.

ANs:

Regression Estimator with Two Auxiliary Variables for population mean

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx}(\bar{X} - \bar{x}) + \hat{\beta}_{yz}(\bar{Z} - \bar{z})$$

$$\hat{\beta}_{yx} = \frac{S_{yx}}{S_x^2}, \hat{\beta}_{yz} = \frac{S_{yz}}{S_z^2}$$

Notations

$$e_0 = (\bar{y} - \bar{Y})/\bar{Y}, e_1 = (\bar{x} - \bar{X})/\bar{X}, e_2 = (\bar{z} - \bar{Z})/\bar{Z}$$

$$E(e_0) = E(e_1) = E(e_2) = 0$$

$$E(e_0^2) = \lambda C_y^2, E(e_1^2) = \lambda C_x^2, E(e_2^2) = \lambda C_z^2, E(e_0 e_1) = \lambda C_{yx}$$

$$\text{where, } C_{yx} = \rho_{yx} C_y C_x$$

Variance of Regression Estimator

$$\bar{y}_{reg} = \bar{y} + \hat{\beta}_{yx}(\bar{X} - \bar{x}) + \hat{\beta}_{yz}(\bar{Z} - \bar{z})$$

mcq's

$$\bar{y}_{reg} = \bar{Y}(1 + e_0) + \hat{\beta}_{yx}(\bar{X} - \bar{X}(1 + e_1)) + \hat{\beta}_{yz}(\bar{Z} - \bar{Z}(1 + e_2))$$

$$\bar{y}_{reg} = \bar{Y}(1 + e_0) - \hat{\beta}_{yx}(\bar{X}e_1) - \hat{\beta}_{yz}(\bar{Z}e_2)$$

$$E(\bar{y}_{reg} - \bar{Y})^2 = E[\bar{Y}e_0 - \hat{\beta}_{yx}(\bar{X}e_1) - \hat{\beta}_{yz}(\bar{Z}e_2)]^2$$

$$V(\bar{y}_{reg}) = E[\bar{Y}^2 e_0^2 + \hat{\beta}_{yx}^2 \bar{X}^2 (e_1)^2 + \hat{\beta}_{yz}^2 \bar{Z}^2 (e_2)^2 - 2\bar{Y}\bar{X}\hat{\beta}_{yx}(e_0e_1) - 2\bar{Y}\bar{Z}\hat{\beta}_{yz}(e_0e_2) + 2\bar{X}\bar{Z}\hat{\beta}_{yz}\hat{\beta}_{yx}(e_1e_2)]$$

$$V(\bar{y}_{reg}) = E[\bar{Y}^2 e_0^2 + \hat{\beta}_{yx}^2 \bar{X}^2 (e_1)^2 + \hat{\beta}_{yz}^2 \bar{Z}^2 (e_2)^2 - 2\bar{Y}\bar{X}\hat{\beta}_{yx}(e_0e_1) - 2\bar{Y}\bar{Z}\hat{\beta}_{yz}(e_0e_2) + 2\bar{X}\bar{Z}\hat{\beta}_{yz}\hat{\beta}_{yx}(e_1e_2)]$$

$$V(\bar{y}_{reg}) = \lambda [\bar{Y}^2 C_y^2 + \hat{\beta}_{yx}^2 \bar{X}^2 C_x^2 + \hat{\beta}_{yz}^2 \bar{Z}^2 C_z^2 - 2\bar{Y}\bar{X}\hat{\beta}_{yx}C_{yx} - 2\bar{Y}\bar{Z}\hat{\beta}_{yz}C_{yz} + 2\bar{X}\bar{Z}\hat{\beta}_{yz}\hat{\beta}_{yx}C_{xz}]$$

$$V(\bar{y}_{reg}) = \lambda S_y^2 [1 - \rho_{yx}^2 - \rho_{yz}^2 - 2\rho_{yx}\rho_{yz}\rho_{xz}]$$

Designs for Hard To Detect Populations

Q. Define Capture Recapture Sampling. imp s/q

Select a sample, mark the selected units and released back to the population.

Select second sample independently from the population.

Estimate the proportion of the marked units from the second sample.

Suppose

T = Number of animals in the population

K = Number of animals marked on the first visit

n = Number of animals captured on the second visit

k = Number of recaptured animals that were marked

imp mcq's

Since the proportion of the marked subjects in the recaptured sample is likely to be about the same as the first sample in the whole population

$$\frac{k}{n} = \frac{K}{T}$$

The estimated population size is

$$T = \frac{nK}{k}$$

Example 1.

When you are interested to estimate the homeless people in your city. An initial Sample of 100 homeless people is selected, marked and released. A sample of size 200 is selected on the second visit and 50 of them were selected in the first visit.

$T=??$, $K=100$, $n=200$, $k=50$

$$T = \frac{nK}{k} = \frac{200 \times 100}{50} = 400$$

Example 2.

In a field study $K = 300$ mice are caught in traps, tagged, and released. A few days later the researchers return to the study area and independently capture $n = 200$ mice, of which they find that $k = 50$ have tags. $T=??$, $K=300$, $n=200$, $k=50$

$$T = \frac{nK}{k} = \frac{200 \times 300}{50}$$

Example 3.

We are interested to estimate the size of a population of turtles in a wildlife preserve. An initial Sample of 20 turtles is selected, marked and released. A sample of size 30 is selected on the second visit and 10 marked turtles found among them.

$$T = \frac{nK}{k} = \frac{20 \times 30}{10} = 60$$

Q. Define Line and Point Transects

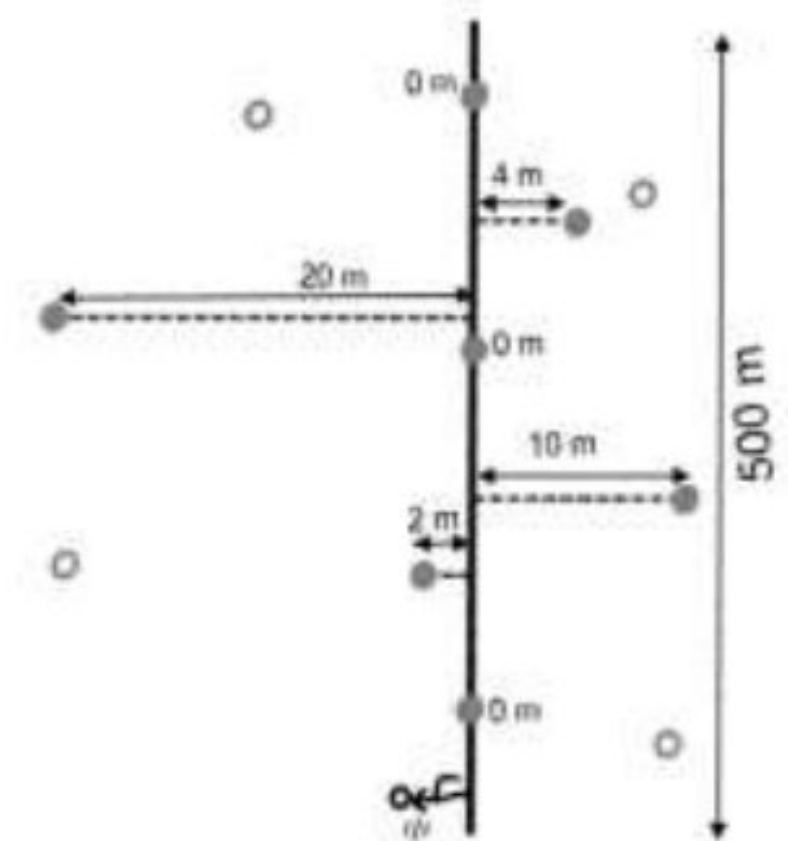
Mostly used for animal or plant species. Select the sample of n lines of size L . Observer moves along a selected line and notes the location relative to the line of every individual of the species detected.

Narrow-Strip Method

$$D = \frac{y_0}{2w_0L}$$

$$Est.Total = AD = \frac{Ay_0}{2w_0L}$$

mcq's



Example

On a line transect of length $L = 100$ meters, a total of $y = 18$ birds were detected at the following distances (in meters) from the transect line

0, 0, 1, 3, 7, 11, 11, 12, 15, 15, 18, 19, 21, 23, 28, 33, 34, 44.

It is desired to estimate the density of birds in the study region. $w_0 = 20$

$$D = \frac{y_0}{2w_0L}$$

Sites are selected as the lines are selected in line transects.

Observations are obtained on selected sites.

Estimation is same as in line transects

163. Explain Adaptive Cluster Sampling

When the selection procedure depends on the observations during the survey

Select an initial sample of size n with a suitable design.

Observe the selected units for a specified condition.

If any of the initially selected unit satisfied the pre-defined condition, its adjacent neighboring units will be sampled and investigated.

Adaptive Cluster Sampling:

- When the selection procedure depends on the observations during the survey.
- Select an initial sample of size n with a suitable design.
- Observe the selected units for a specified condition.
- If any of the initially selected unit satisfied the pre-defined condition, its adjacent neighboring units will be sampled and investigated.

The sample mean is

$$\bar{w}_y = \frac{1}{n} \sum_{i=1}^n w_{yi} \quad \text{imp s/q}$$

All MSE Error Question important and also biase for long and short and do practice for R language code of HH, HT etc.

Regards: Jannat Rana

Final Term

End

Regard:JannatRana